



UNIMORE

UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Dipartimento di Ingegneria
“Enzo Ferrari”

Fondamenti di Elettronica

Nuts and Bolts of Electronics

Corso di Laurea in Ingegneria Informatica

Prof. Francesco Maria Puglisi
email: francescomaria.puglisi@unimore.it
phone: +39-059-2056352

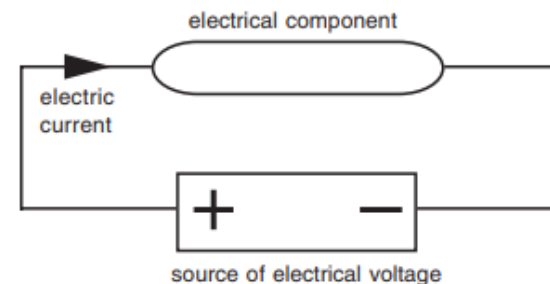
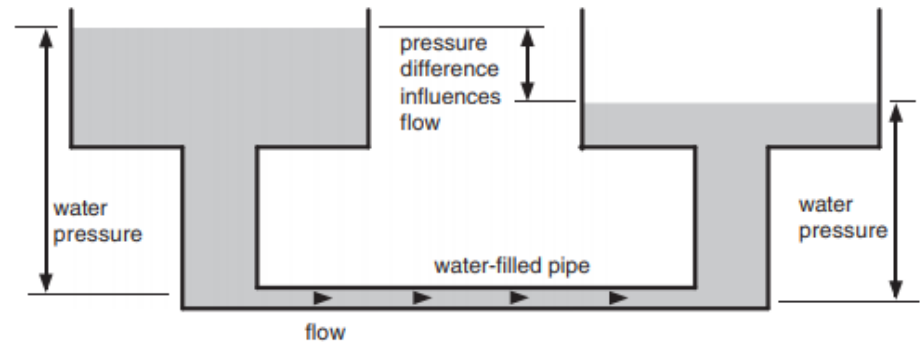
Voltage and Current

The Main Characters

A circuit comprises **a number of interconnected parts**, each of which imposes **its own unique relationship** between two electrical quantities: **voltage (V)** and **current (I)**.

It is these relationships, and their variety, which makes it possible for the creative circuit designer to design many different useful circuits.

Once we know how the parts are connected to form a circuit we can then describe that circuit (essentially a new component) in terms of the relationships that it now imposes upon its voltages and currents.



Voltage and Current Conventions

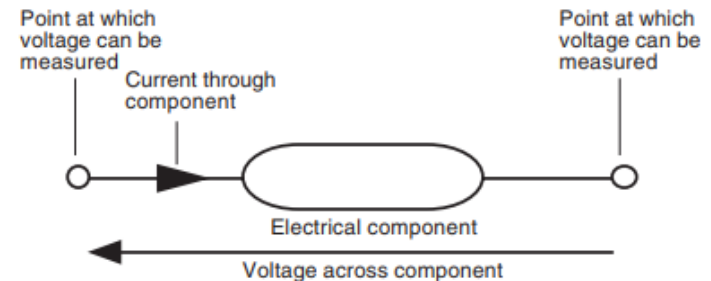
A circuit comprises **a number of interconnected parts**, each of which imposes **its own unique relationship** between two electrical quantities: **voltage (V)** and **current (I)**.

Voltage is measured in **Volt**

Current is measured in **Ampere** (i.e., C/s)

Arrows are used to indicate voltages and currents. An arrow alongside a **circuit branch** indicates the current flow direction (by convention opposed to the electrons flow). A stretching arrow between two **circuit nodes** indicates the voltage difference between those nodes.

Another important player: **Power** (measured in **Watts**). For some components, V is always positive if I is positive, so energy can only be supplied to these components. These are **passive** components. The energy supplied to passive components is dissipated as **heat**.



$$[V] = V$$

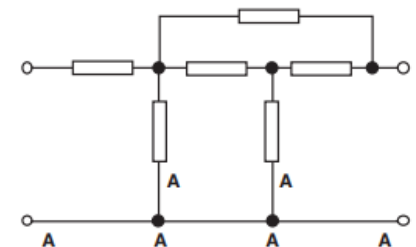
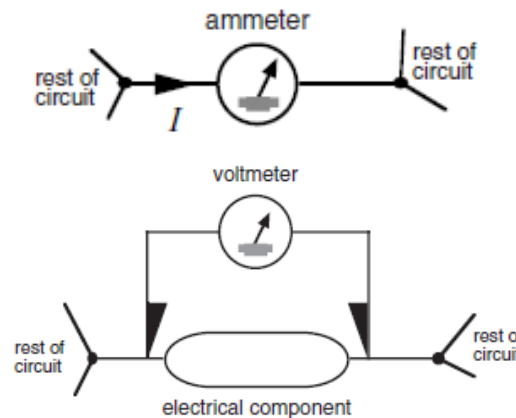
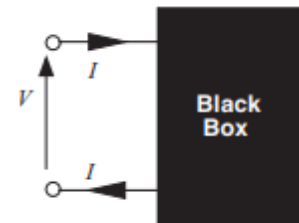
$$[I] = A = C/s$$

$$[Q] = C$$

$$[P] = W = J/s = V \cdot A$$

$$[E] = J$$

$$P = V \cdot I$$



Basic Components

Open and Short Circuits



Short Circuit

$V = 0$ regardless the current flow

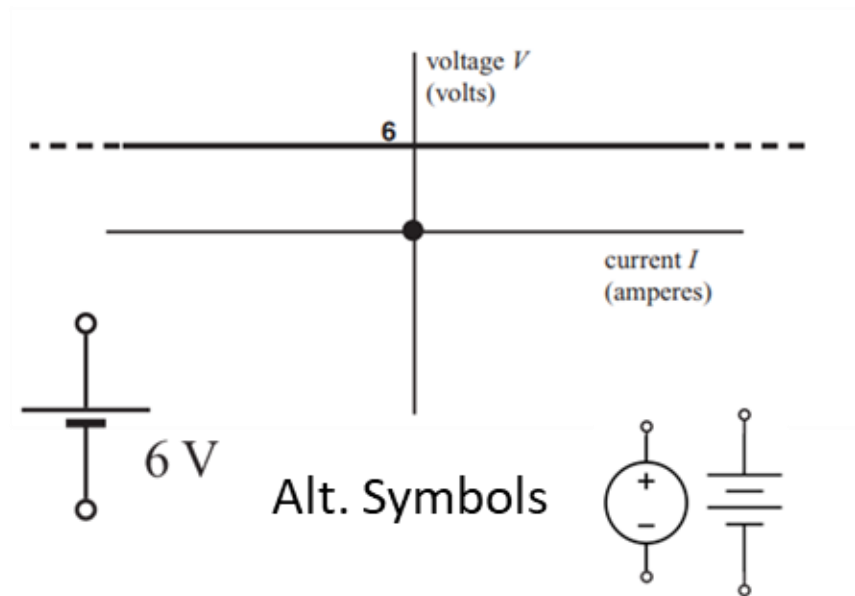


Open Circuit

$I = 0$ regardless the voltage across

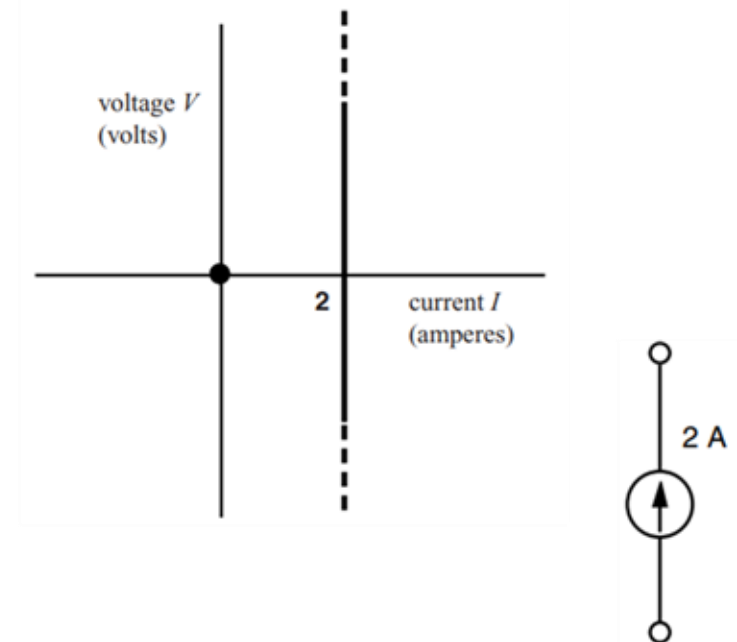
Basic Components

Ideal Sources



Ideal Voltage Source

V constant regardless the current flow
(current can flow in any direction)



Ideal Current Source

I constant regardless the voltage across
(voltage can be of any polarity)

Basic Components

Resistor

Constitutive relation of a resistor
(a.k.a Ohm's Law)

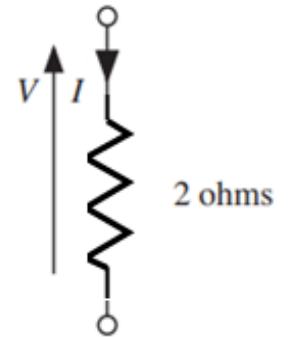
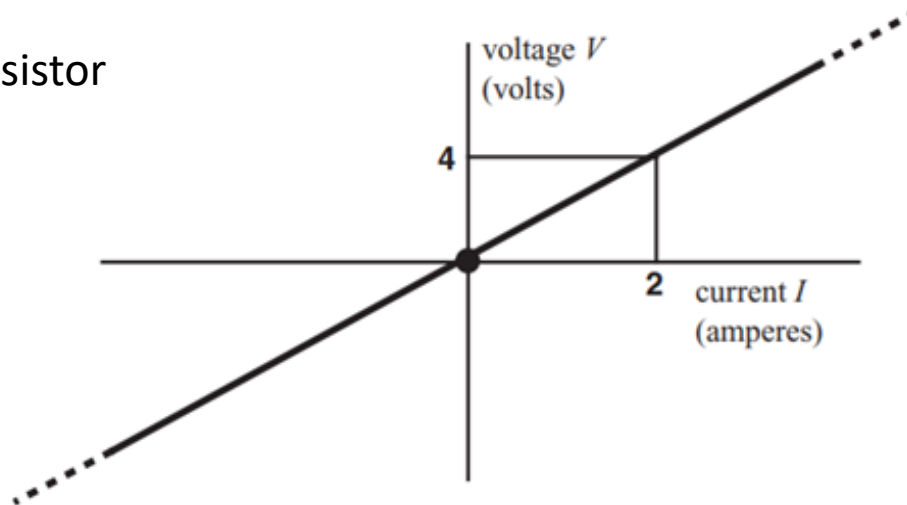
$$V = R \cdot I$$

R is the Resistance
[R] = Ω (Ohm)

$$R = G^{-1}$$

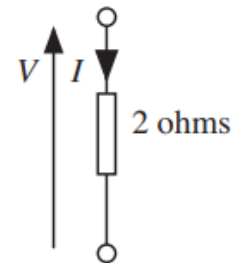
$$I = G \cdot V$$

G is the Conductance
[G] = $\Omega^{-1} = S$ (Siemens)



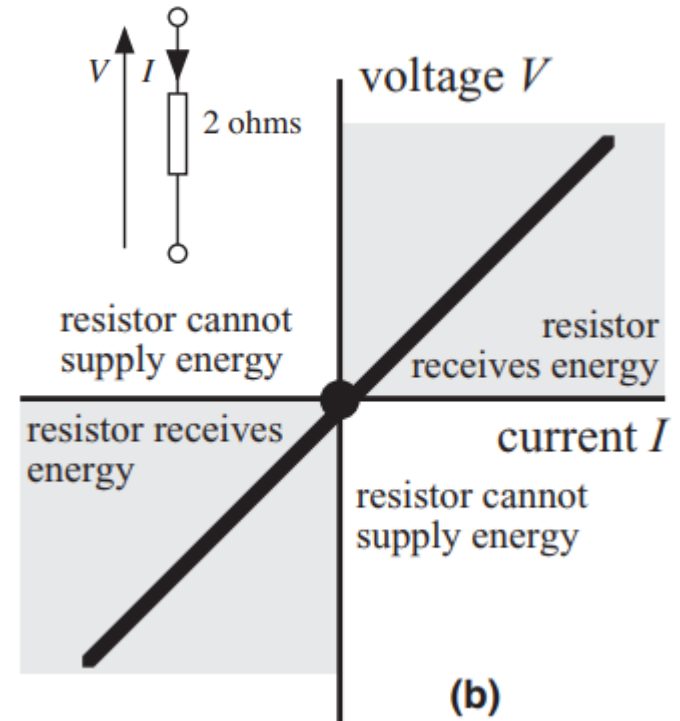
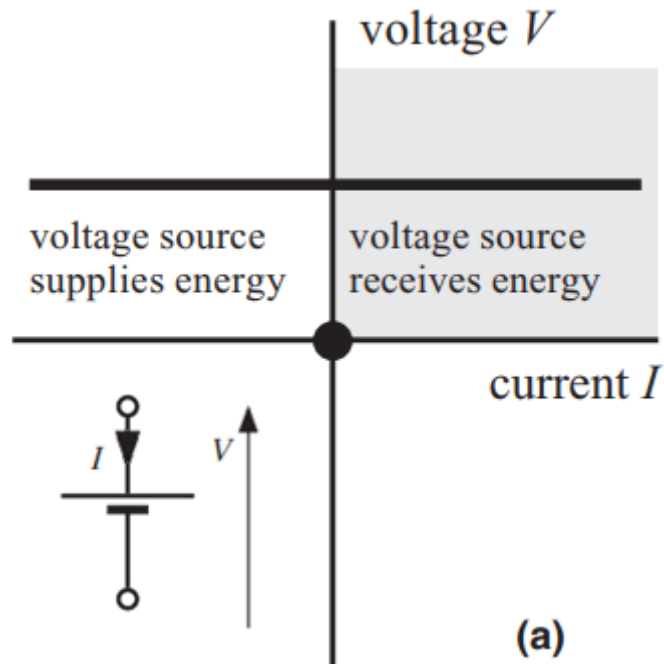
Resistor

V is proportional to I
(+ve voltage implies +ve current)
(-ve voltage implies -ve current)



Basic Components

Power

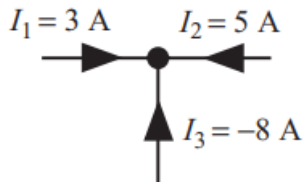
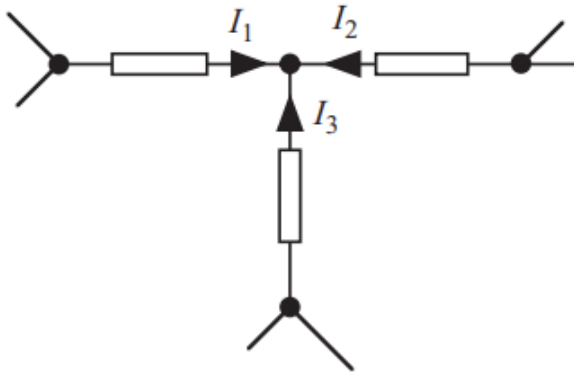


$$P = V \cdot I = R \cdot I^2 = V^2/R$$

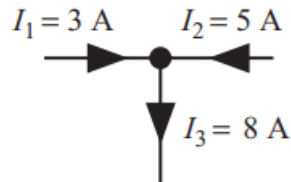
Basic Components

Kirchhoff's Laws

KCL



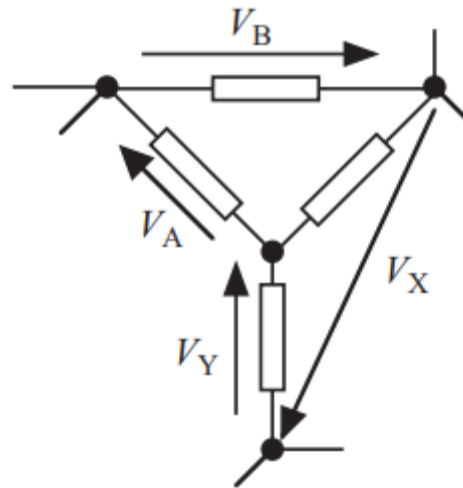
"Total current into a node is zero"



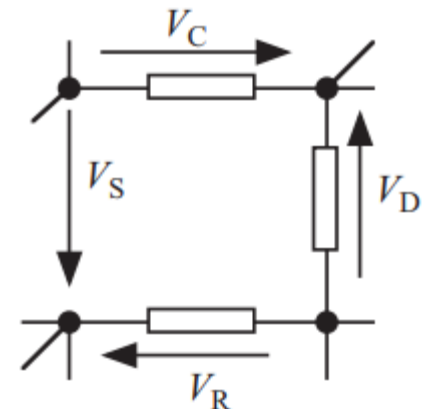
"Total current in equals total current out"

$$\sum_{\text{node}} I = 0$$

KVL



$$V_A + V_B + V_X + V_Y = 0$$



$$V_C - V_D + V_R - V_S = 0$$

$$\sum_{\text{loop}} V = 0$$

Basic Components

Resistors: Series and Parallel Connections

$$V = V_1 + V_2$$

$$I = I_1 = I_2$$

$$V_1 = R_1 \cdot I_1 = R_1 \cdot I$$

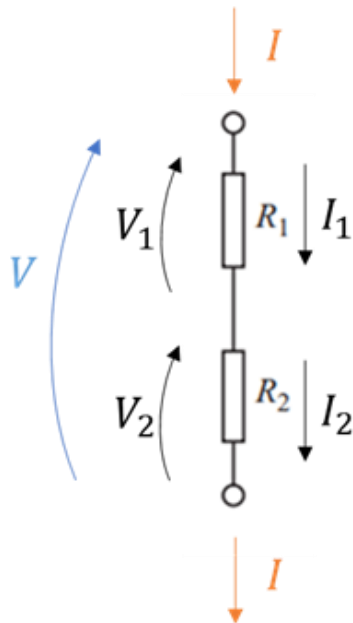
$$V_2 = R_2 \cdot I_2 = R_2 \cdot I$$

$$V = (R_1 + R_2) \cdot I$$

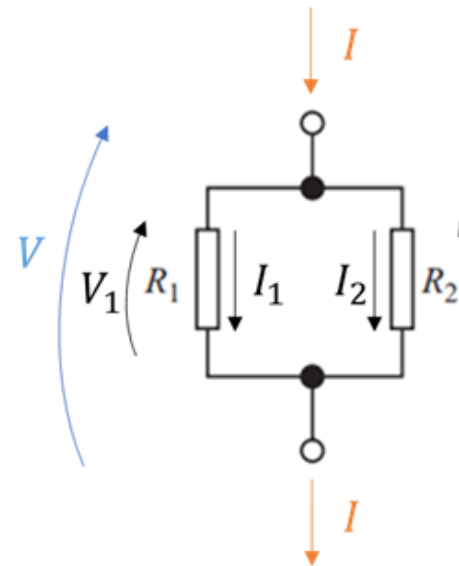
$$V = R_{eq} \cdot I$$

$$R_{eq} = R_1 + R_2$$

Series Connection



Parallel Connection



$$V = V_1 = V_2$$

$$I = I_1 + I_2$$

$$I_1 = G_1 \cdot V_1 = G_1 \cdot V$$

$$I_2 = G_2 \cdot V_2 = G_2 \cdot V$$

$$I = (G_1 + G_2) \cdot V$$

$$I = G_{eq} \cdot V$$

$$G_{eq} = G_1 + G_2$$

$$R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 \cdot R_2}{R_1 + R_2} = R_1 \parallel R_2$$

Series Connection (Voltage Divider)

Same current flowing into both resistors

Voltage across the two resistor **may be different**

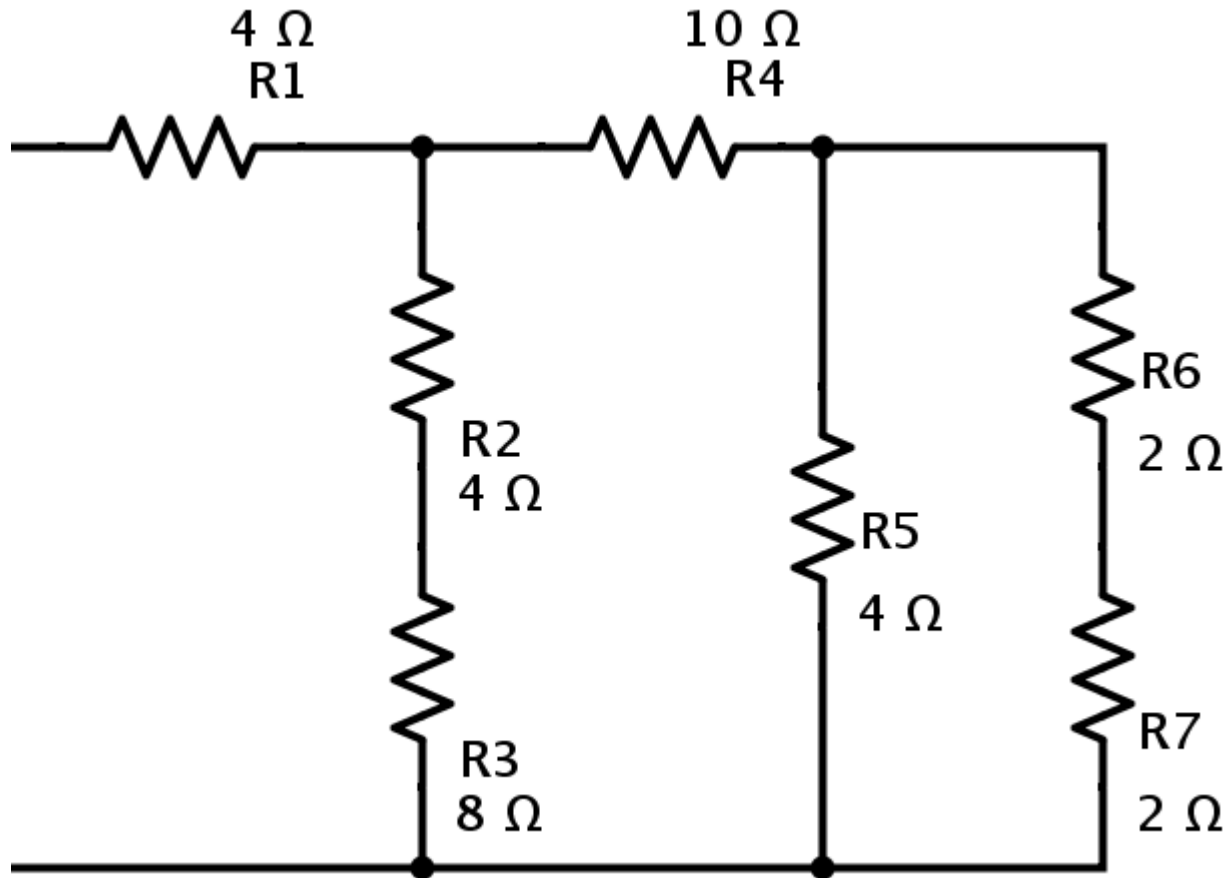
Parallel Connection (Current Divider)

Same voltage across resistors

Current on the two resistor **may be different**

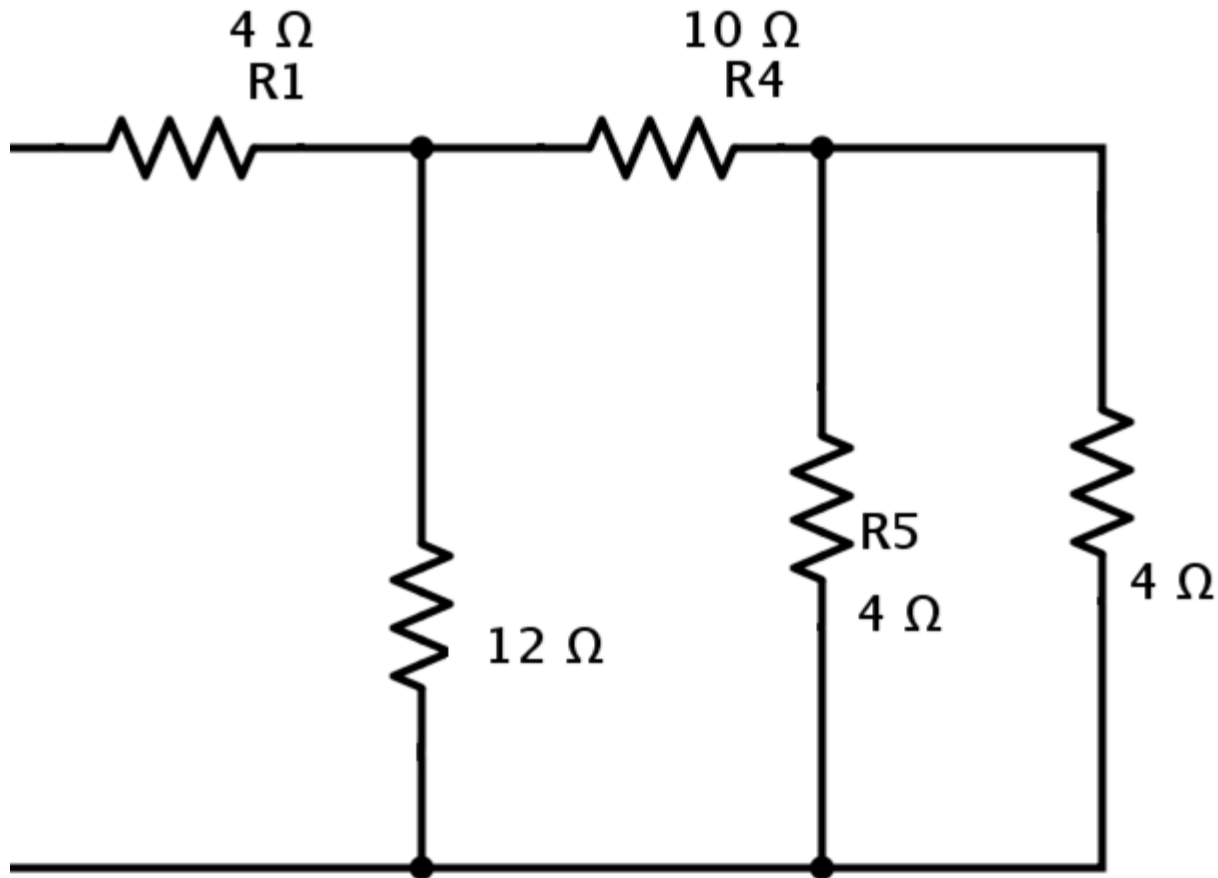
Basic Components

Exercise on a Simple Circuit



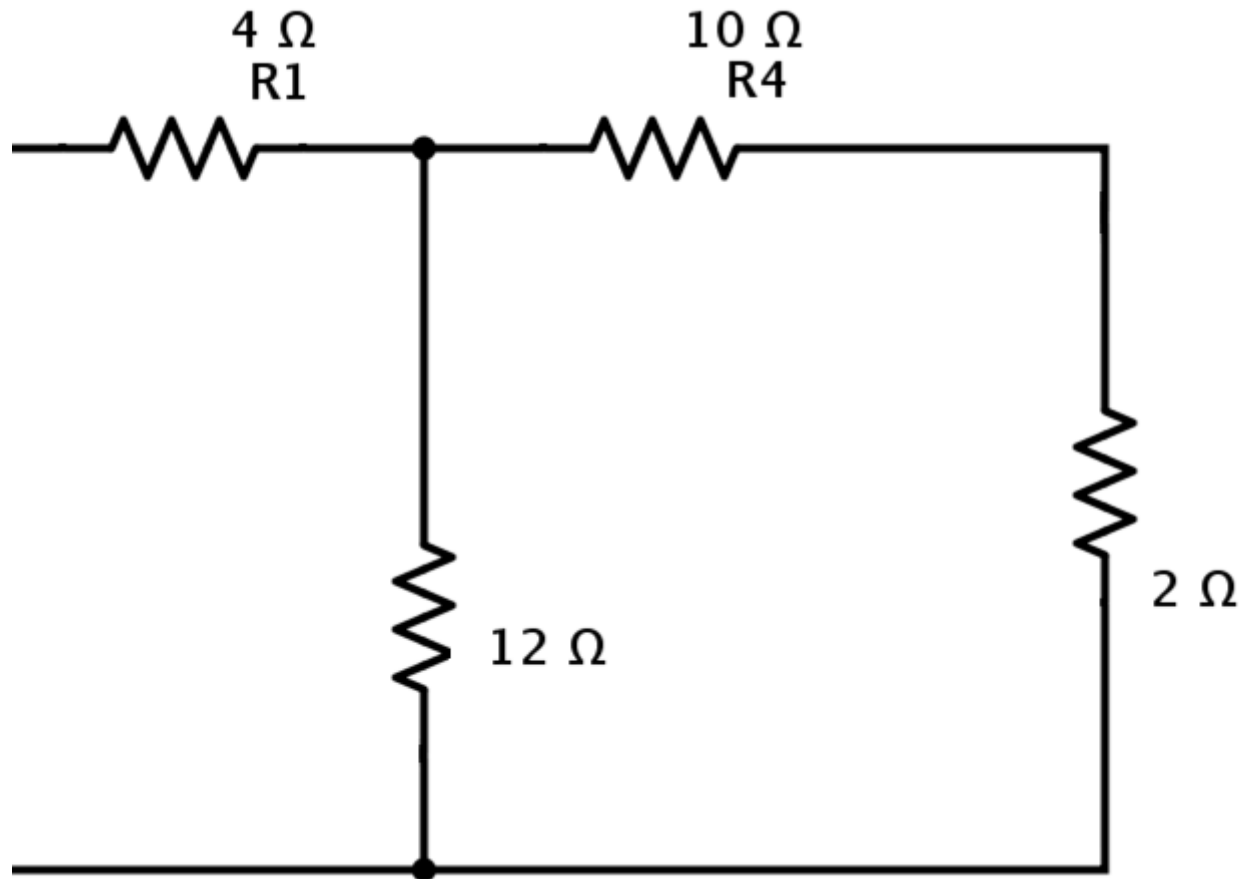
Basic Components

Exercise on a Simple Circuit



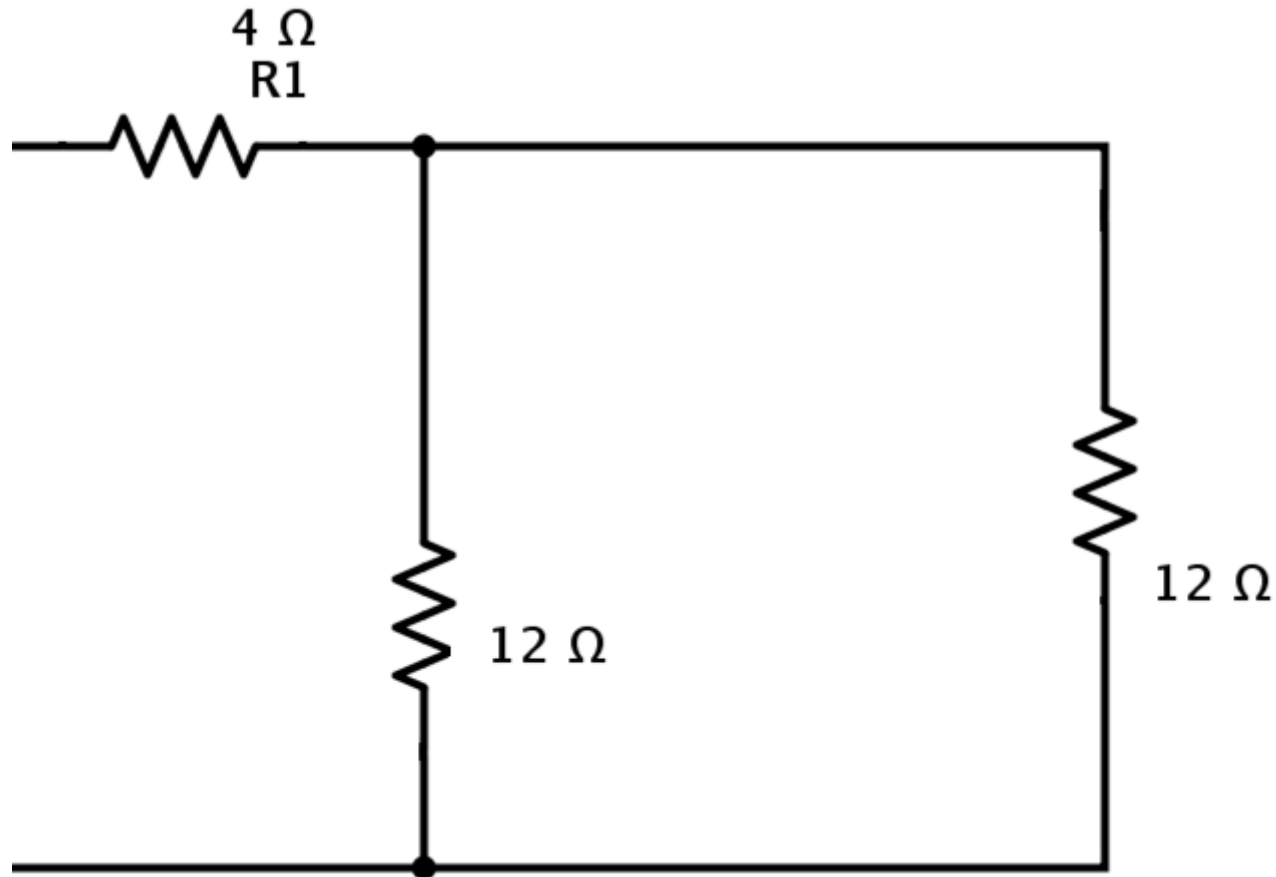
Basic Components

Exercise on a Simple Circuit



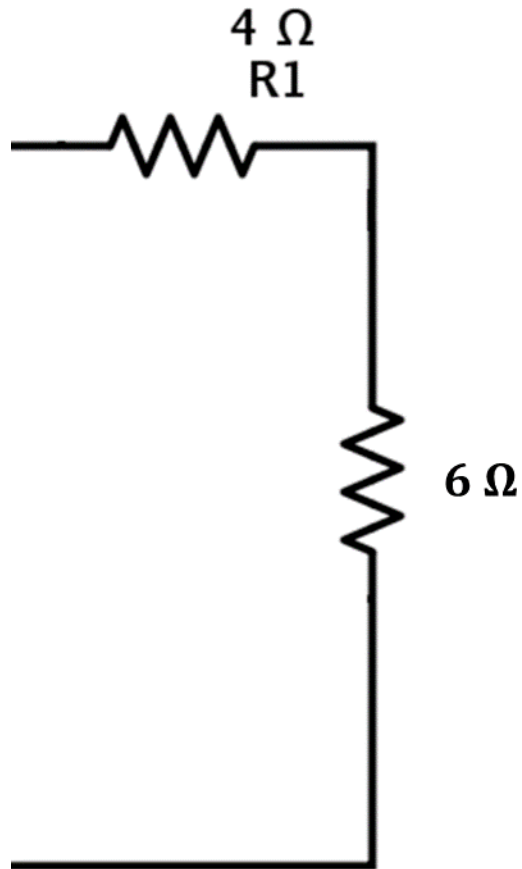
Basic Components

Exercise on a Simple Circuit



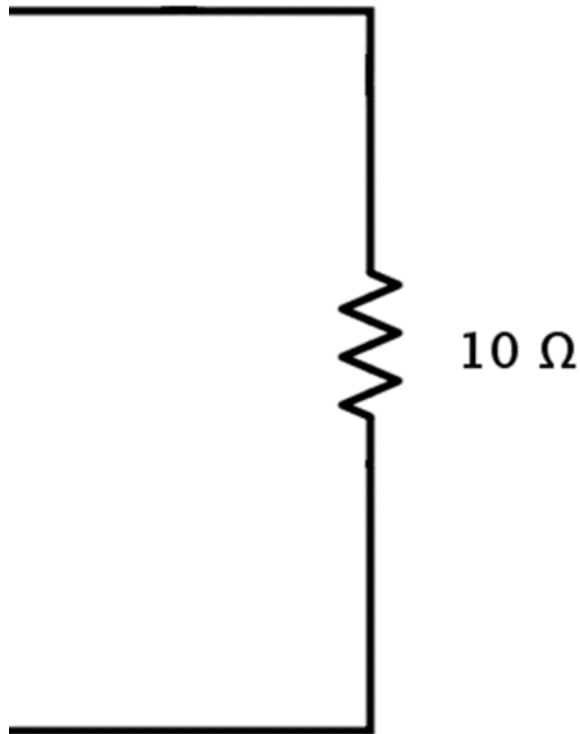
Basic Components

Exercise on a Simple Circuit



Basic Components

Exercise on a Simple Circuit



Basic Components

Capacitor

Constitutive relation of a capacitor

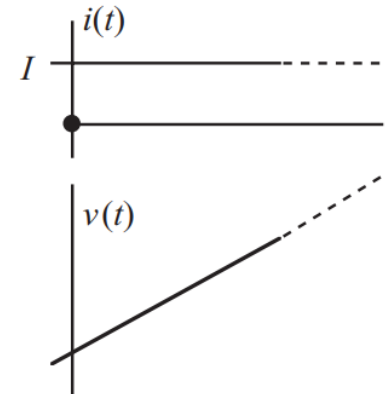
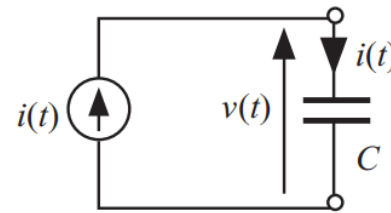
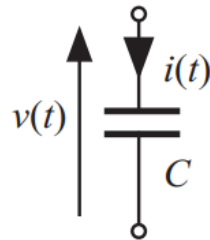
$$Q = C \cdot V$$

Q is the Electric Charge

[Q] = C (Coulomb)

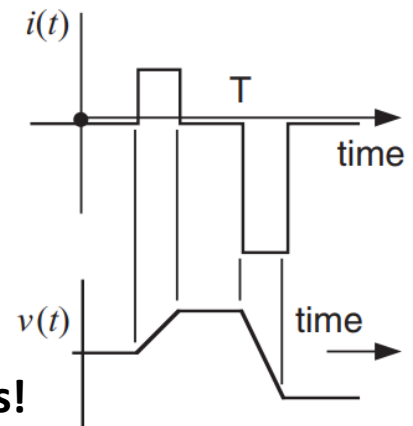
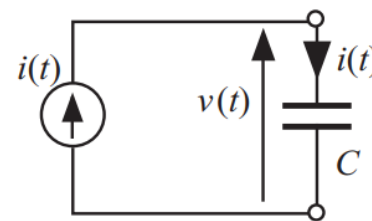
C is the Capacitance

[C] = F (Farad)



Since $I = dQ/dt$ by definition, then

$$I = C \, dV/dt$$



Challenge! Work out the formulae for **series** and **parallel** capacitors!

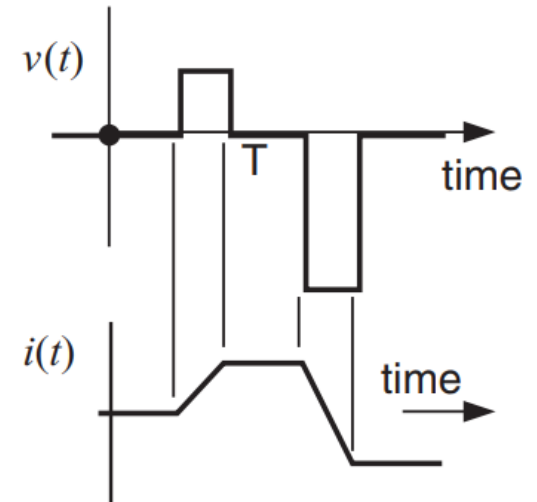
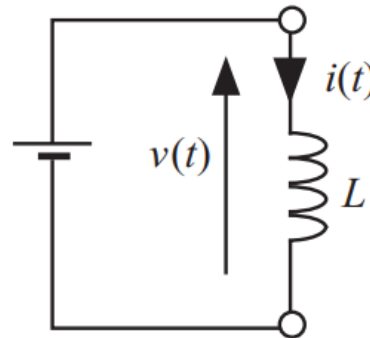
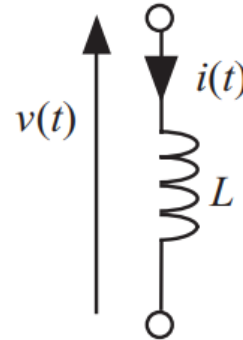
Basic Components

Inductor

Constitutive relation of an inductor

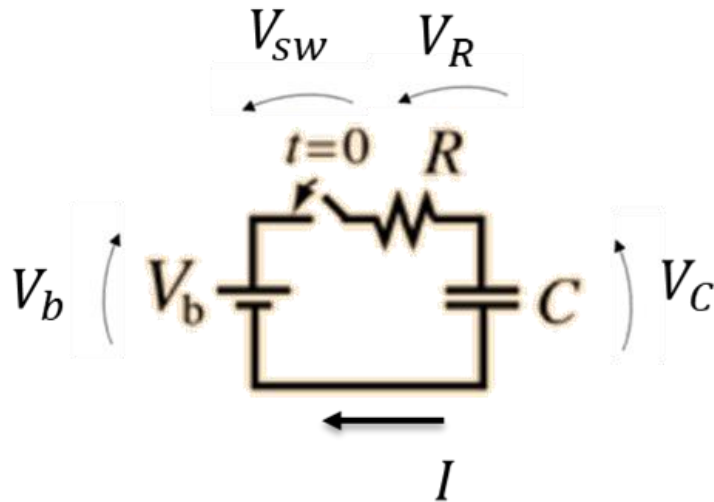
$$V = L \cdot di/dt$$

L is the Inductance
[L] = H (Henry)



Basic Components

Charging and Discharging a Capacitor

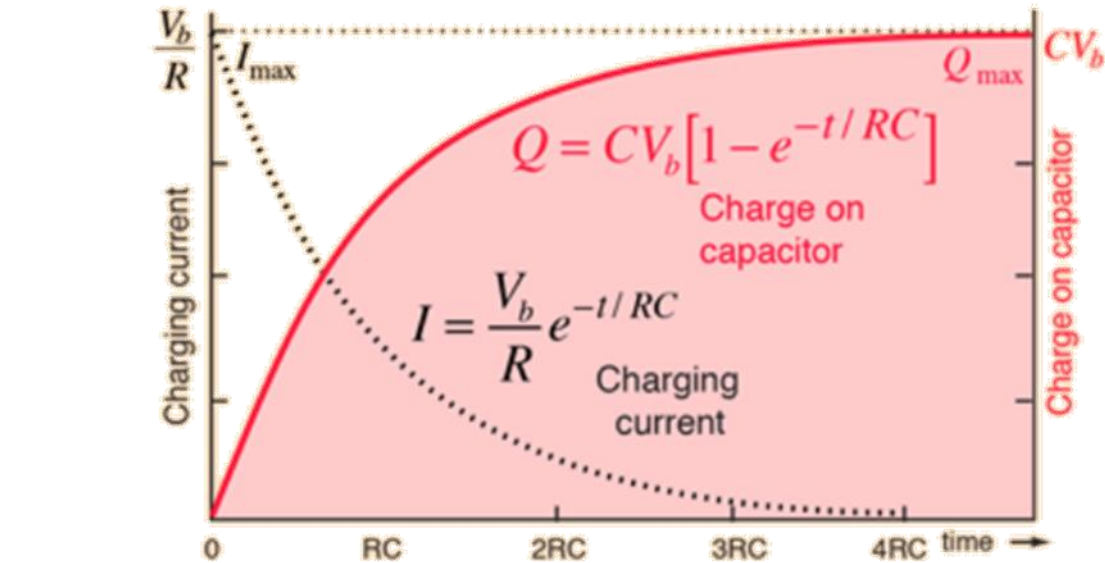


$$V_b = V_{sw} + V_R + V_C$$

$$V_b = V_{sw} + IR + \frac{Q}{C} = V_{sw} + IR + \frac{\int I dt}{C}$$

$$\text{for } t < 0 \rightarrow V_b = V_{sw} + \frac{Q}{C}$$

$$\text{for } t \geq 0 \rightarrow V_b = IR + \frac{Q}{C}$$



$V_b = IR + \frac{Q}{C}$

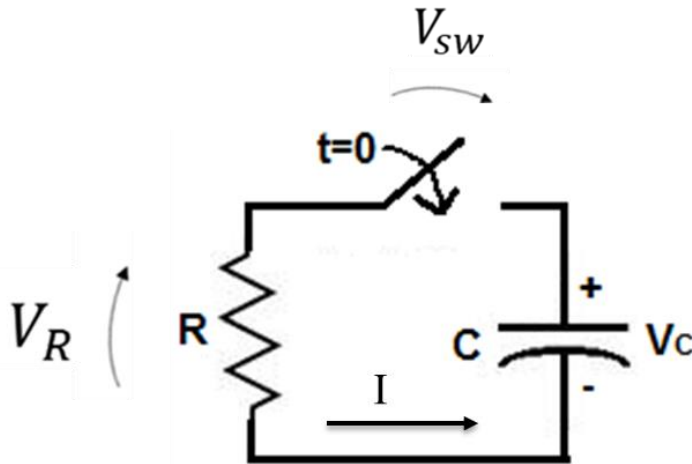
 current decreases and charge increases.

At $t = 0$	As $t \rightarrow \infty$
$Q = 0$	$Q \rightarrow CV_b$
$V_C = 0$	$V_C \rightarrow V_b$
$I = \frac{V_b}{R}$	$I \rightarrow 0$

Basic Components

Charging and Discharging a Capacitor

What happens to a circuit like this one if the capacitor is **pre-charged to a voltage V_C** ?



$$V_R + V_{sw} = V_C$$

$$\begin{array}{ll} \text{for } t < 0 & V_{sw} = V_C \\ \text{for } t \geq 0 & V_R = V_C \end{array}$$

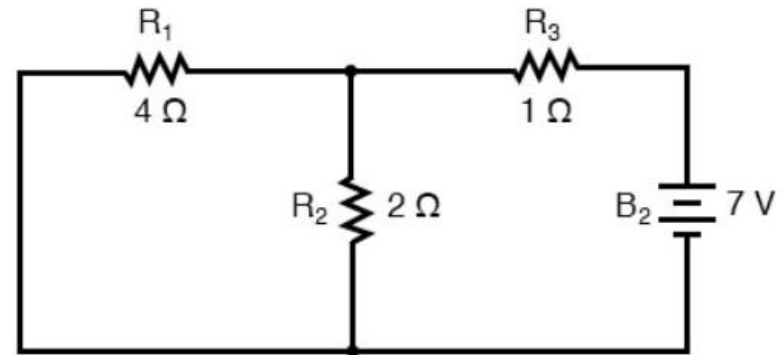
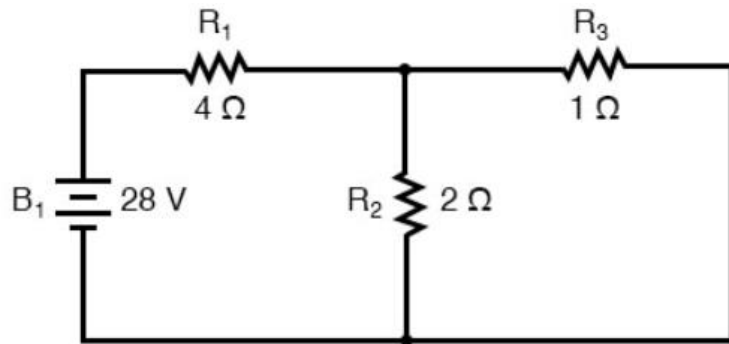
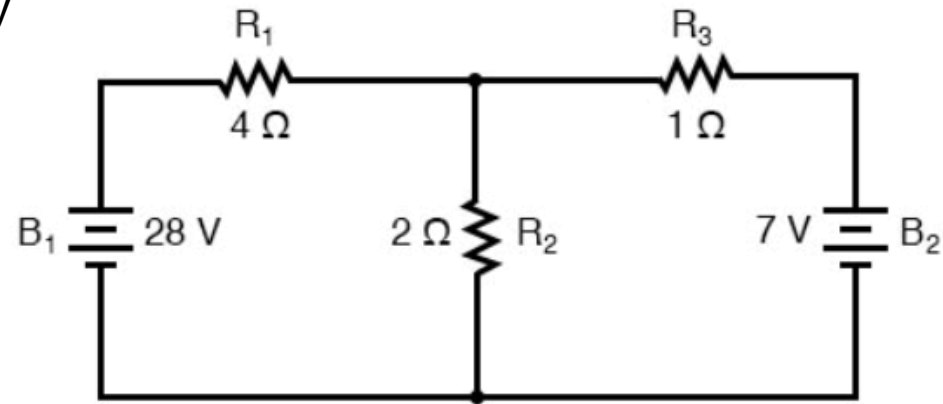
Can you work out the **time evolution** of the current and the voltages in the circuit?

Superposition Principle

More than one Source

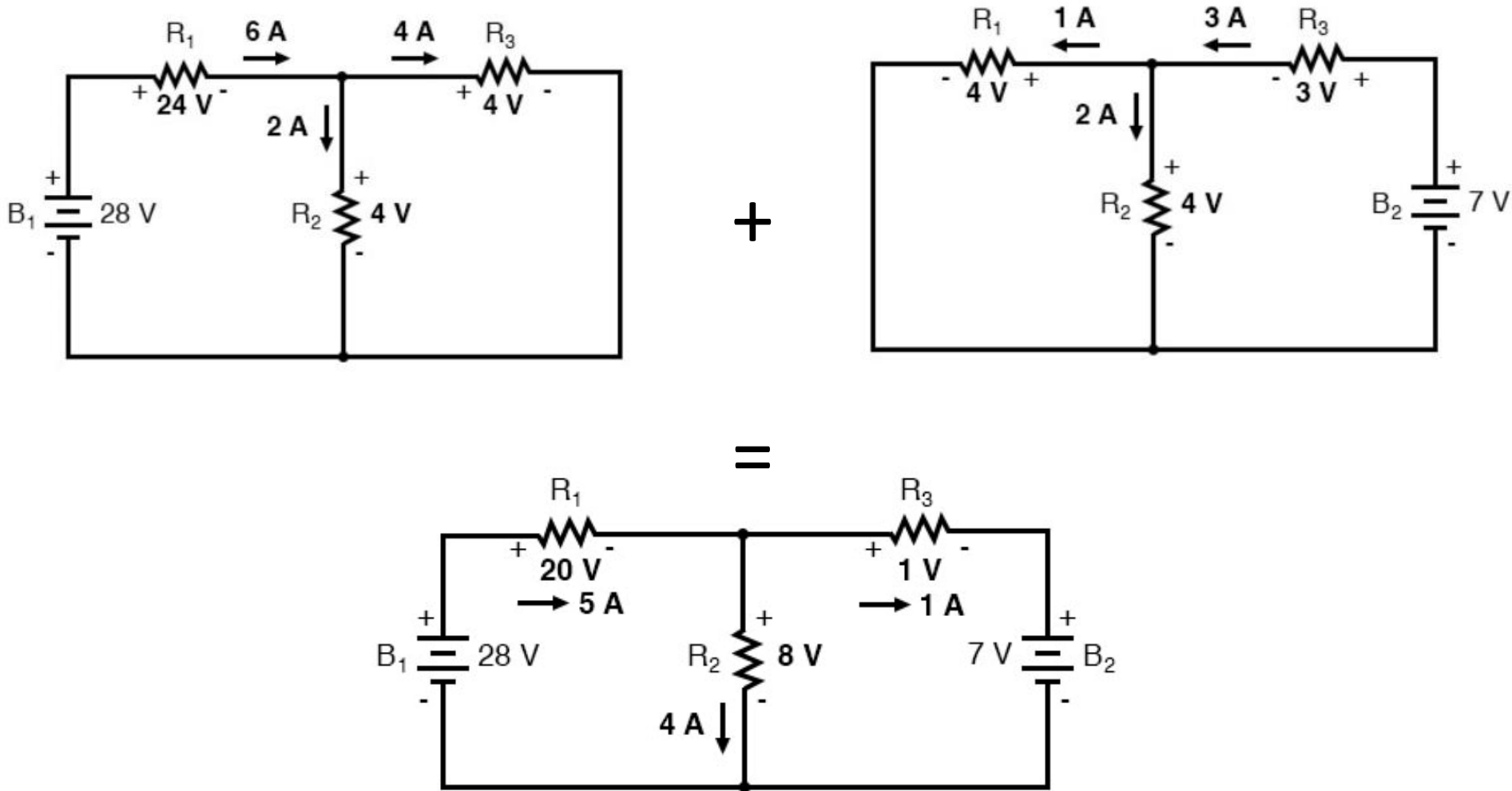
When **more than one independent source** is present in a linear circuit, we can apply the **superposition principle**

1. Draw n equivalent circuits (one per independent source) by “keeping” only one source per equivalent circuit
2. Superpose their effects



Superposition Principle

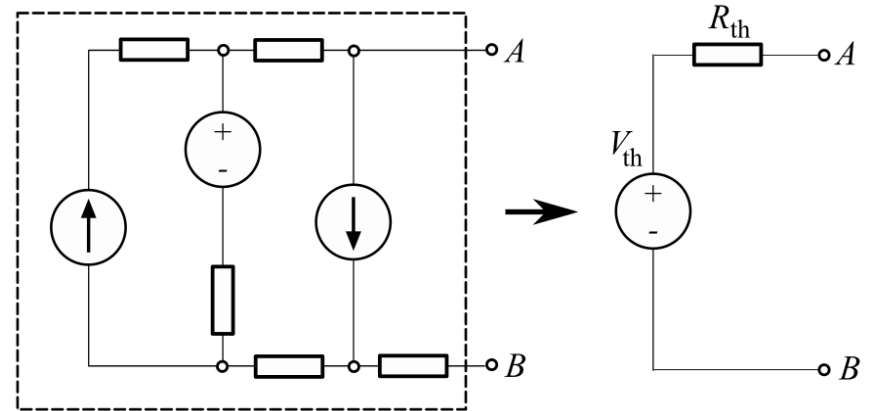
More than one Source



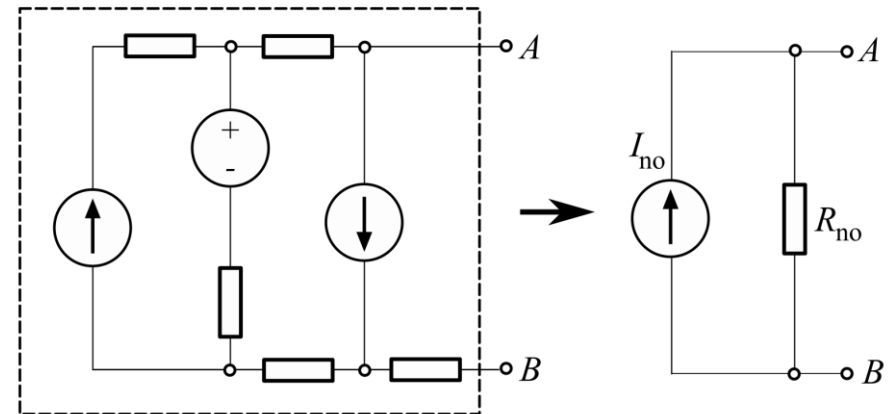
Equivalent Circuits

Thevenin and Norton

Thevenin's theorem states that a **generic black box non-dynamic linear circuit** observed by two nodes can be represented by a **voltage source in series with a resistor**.



Norton's theorem states that a **generic black box non-dynamic linear circuit** observed by two nodes can be represented by a **current source in parallel with a resistor**.



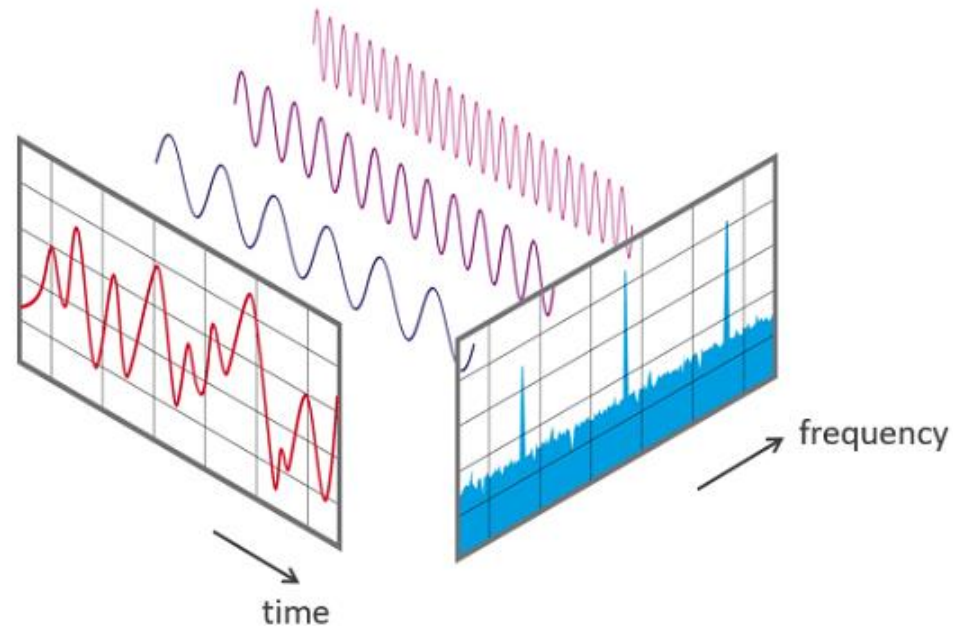
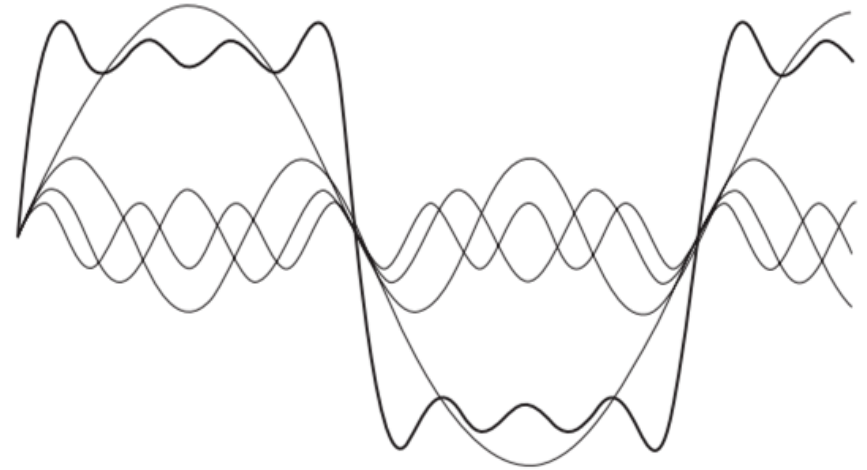
Math of Signals

DC vs. AC

DC means Direct Current (constant voltages and current in the whole circuit)

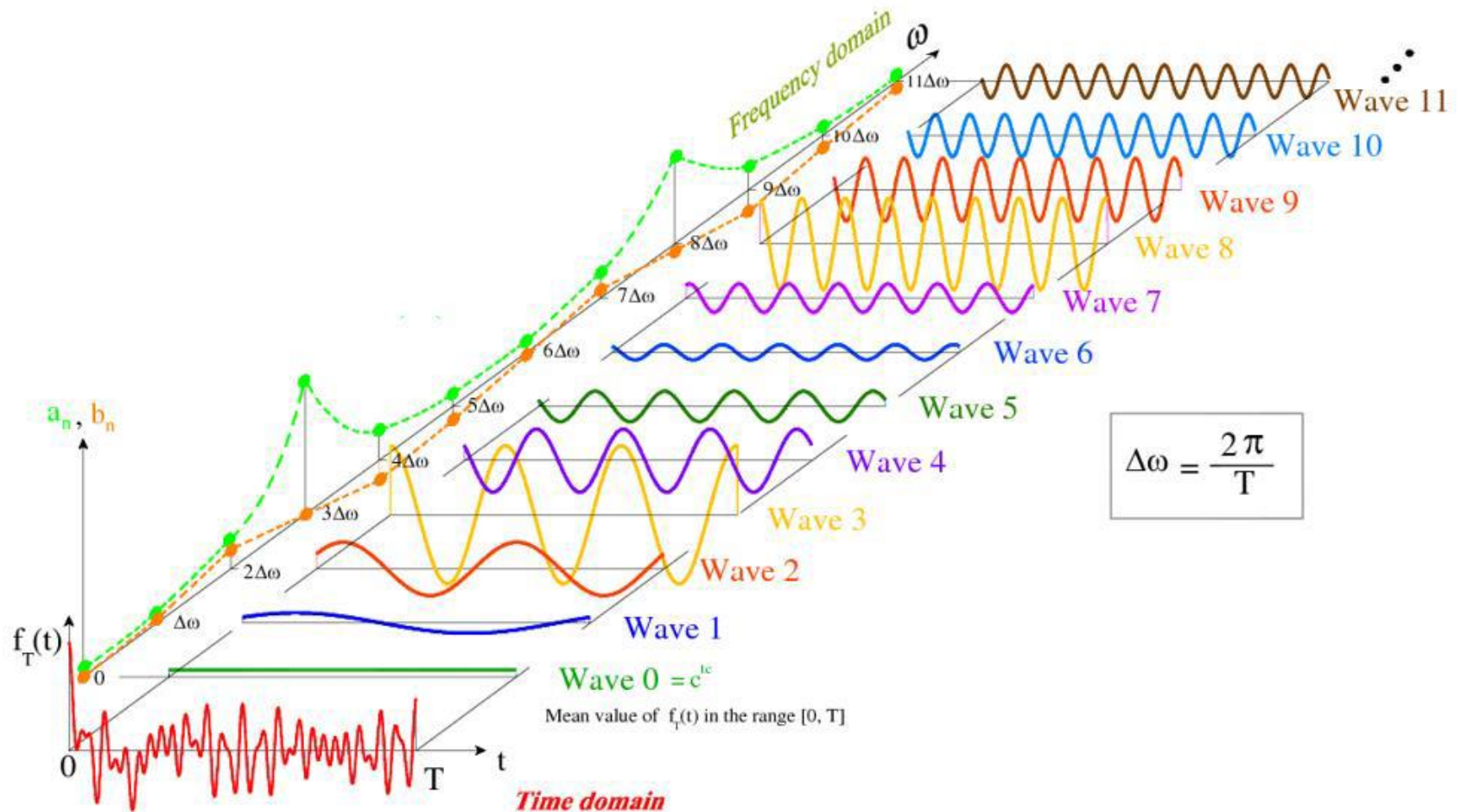
AC means Alternating Current (voltage and current varying in the circuit). Every AC signal can be seen as a superposition of sine waves (Fourier Theorem). Important to study the response of the system to sine waves at any frequency!

Reactive components like capacitors and inductors only respond to AC (and transient) signals!



Math of Signals

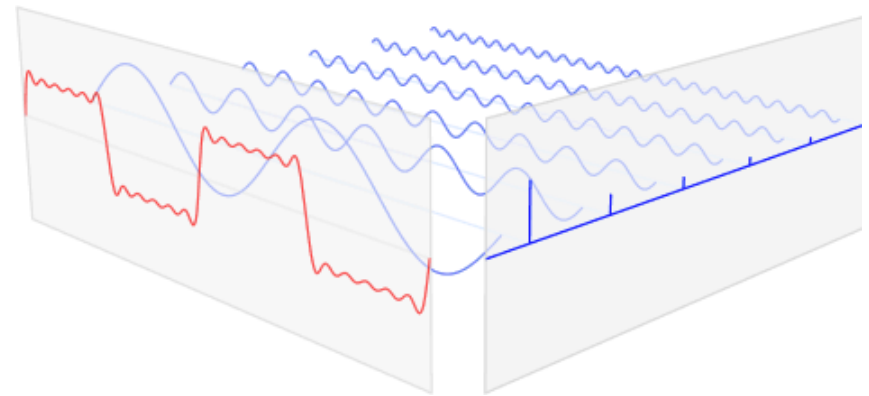
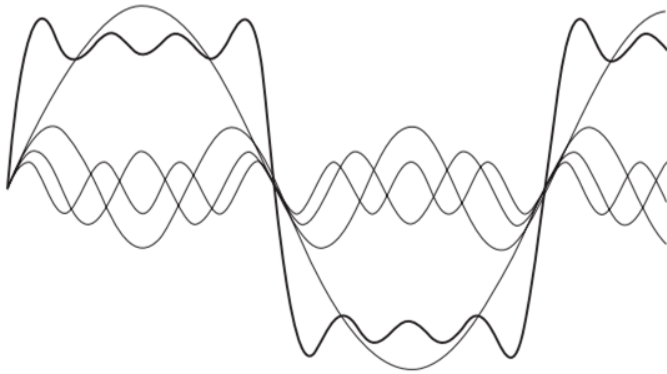
DC vs. AC



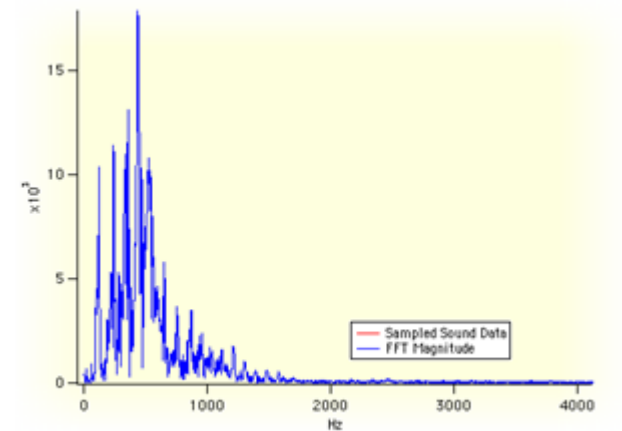
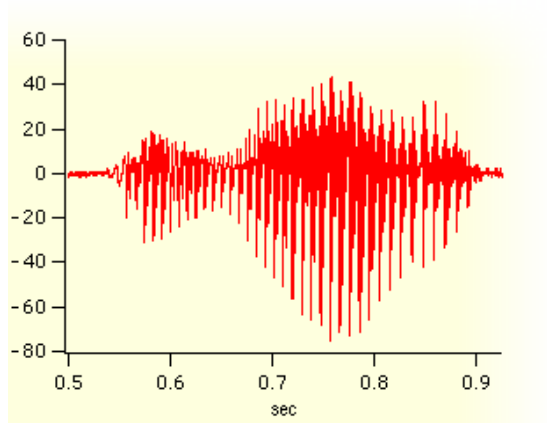
Math of Signals

DC vs. AC

Periodic AC signals: Discrete Spectrum



Non-Periodic AC signals: Continuous Spectrum



Math of Signals

A Tale of Two Domains

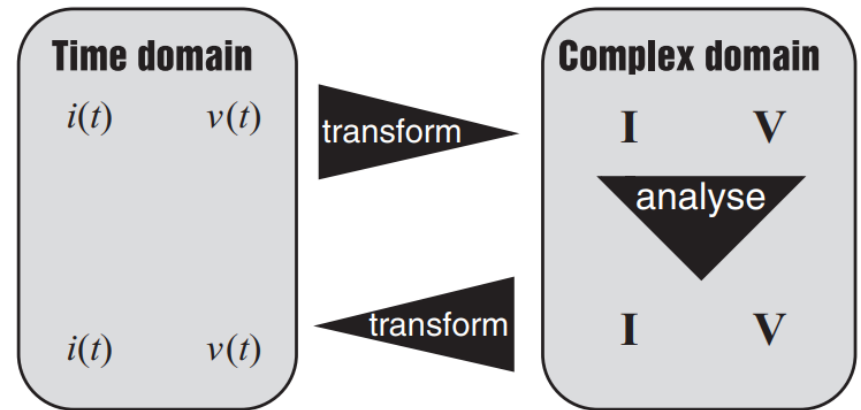
Time representation involves functions of time like sine and cosine.

Phasor representation involves rotating vectors with fixed amplitude (phasors) represented by complex numbers. Extremely useful in computing the response of circuits to AC signals, as compared to the time representation. Why? It is so much easier!

Can easily go back and forth between the two representations.

We will study:

- Phasor representation
- Impedance
- Series and Parallel connection of resistive and reactive items



Recall «Euler's Theorem»

$$e^{j\theta} = \cos \theta + j \sin \theta$$
$$e^{j\omega t} = \cos \omega t + j \sin \omega t$$

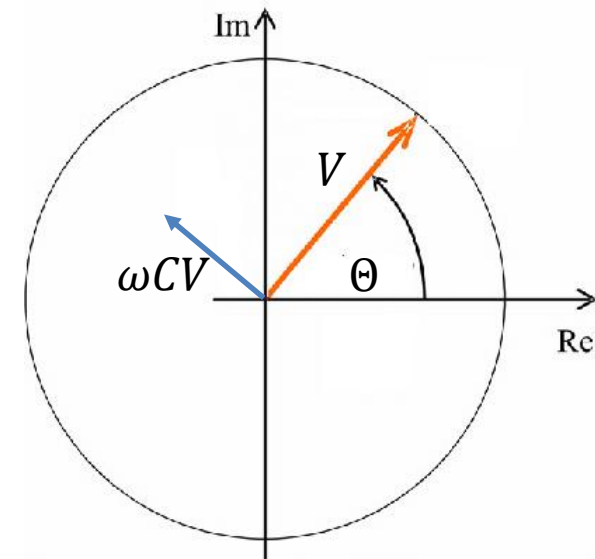
Apply it to sin or cos waveforms

$$v(t) = V \cos(\omega t + \theta)$$

$$v(t) = \text{Re}[V e^{j(\omega t + \theta)}] = \text{Re}[V e^{j\theta} e^{j\omega t}]$$

Math of Signals

A Tale of Two Domains



Recall Euler's Theorem:

$$e^{j\omega t} = \cos \omega t + j \cdot \sin \omega t$$

Apply it to sine or cosine waveforms:

$$v(t) = V \cdot \cos(\omega t + \theta) \rightarrow v(t) = \text{Re}[V \cdot e^{j(\omega t + \theta)}] = \text{Re}[V \cdot e^{j\theta} \cdot e^{j\omega t}]$$

- Calculate the impedance of a capacitor:

$$v(t) = V \cdot \cos(\omega t + \theta) = \text{Re}\{V \cdot e^{j\theta} \cdot e^{j\omega t}\} = \text{Re}\{\tilde{V} \cdot e^{j\omega t}\}$$

$$\begin{aligned} i(t) &= C \cdot \frac{dv(t)}{dt} = -C \cdot V \cdot \omega \cdot \sin(\omega t + \theta) = C \cdot V \cdot \omega \cdot \cos\left(\omega t + \theta + \frac{\pi}{2}\right) \\ &= \text{Re}\left\{C \cdot V \cdot \omega \cdot e^{j\theta} \cdot e^{j\omega t} \cdot e^{j\frac{\pi}{2}}\right\} \end{aligned}$$

$$\text{since } e^{j\frac{\pi}{2}} = j \text{ and } V \cdot e^{j\theta} = \tilde{V} \quad i(t) = \text{Re}\{j \cdot \omega \cdot C \cdot \tilde{V} \cdot e^{j\omega t}\}$$

$$\text{if } i(t) = \text{Re}\{\tilde{I} \cdot e^{j\omega t}\} \rightarrow \tilde{I} = j \cdot \omega \cdot C \cdot \tilde{V} \text{ since } \tilde{V} = Z\tilde{I} \rightarrow Z_C = \frac{1}{j\omega C} = -j \frac{1}{\omega C}$$

Math of Signals

A Tale of Two Domains

$$i(t) = I \cdot \cos(\omega t + \theta) = \text{Re}\{I \cdot e^{j\theta} \cdot e^{j\omega t}\} = \text{Re}\{\tilde{I} \cdot e^{j\omega t}\}$$

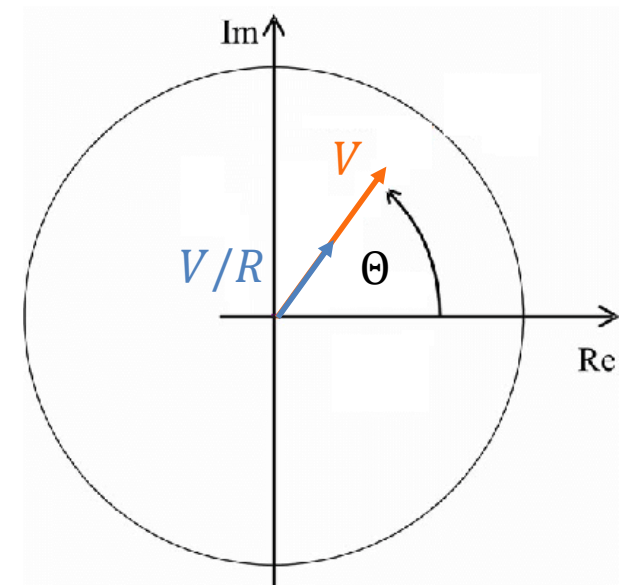
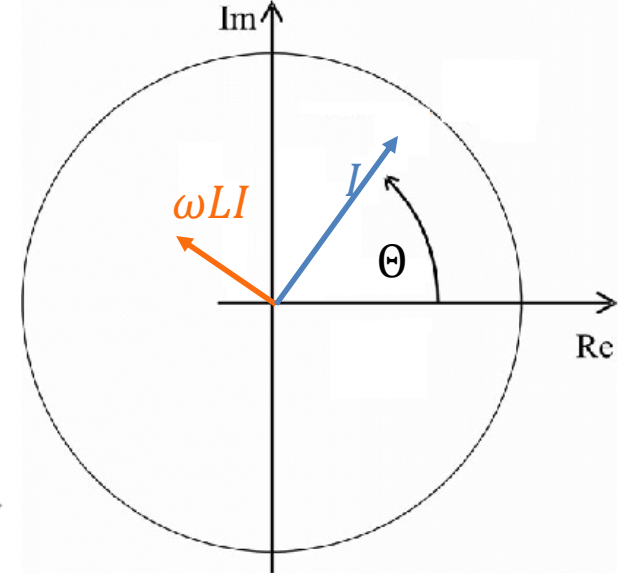
$$v(t) = -L \cdot I \cdot \omega \cdot \sin(\omega t + \theta) = \omega \cdot L \cdot I \cdot \cos\left(\omega t + \theta + \frac{\pi}{2}\right) = \text{Re}\{j \cdot \omega \cdot L \cdot I \cdot e^{j\theta} \cdot e^{j\omega t}\}$$

$$\tilde{V} = j\omega L \cdot \tilde{I} \quad \text{since} \quad \tilde{V} = Z\tilde{I} \rightarrow Z_L = j\omega L$$

$$v(t) = V \cdot \cos(\omega t + \theta) = \text{Re}\{\tilde{V} \cdot e^{j\omega t}\}$$

$$i(t) = \frac{v(t)}{R} = \text{Re}\left\{\frac{\tilde{V}}{R} \cdot e^{j\omega t}\right\}$$

$$\tilde{I} = \frac{\tilde{V}}{R} \quad \text{since} \quad \tilde{V} = Z\tilde{I} \rightarrow Z_R = R$$



Math of Signals

A Tale of Two Domains

$$Z = R + jX \quad \text{where} \quad \begin{cases} R = \text{resistance} & , \quad [R] = \Omega \\ X = \text{reactance} & , \quad [X] = \Omega \end{cases}$$

Remembering that the conductance is defined as:

$$G \equiv \frac{1}{R} \quad , \quad [G] = S \equiv \frac{1}{\Omega} = \Omega^{-1}$$

The same idea can be applied to the admittance:

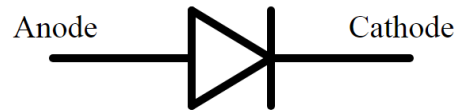
$$Y \equiv \frac{1}{Z} = G + jB \quad \text{where} \quad \begin{cases} G = \text{conductance} & , \quad [G] = S \\ B = \text{susceptance} & , \quad [B] = S \end{cases}$$

To recap, the relations between V and I found for C, L and R allow us to use Ohm's law, with impedances Z, which are in general complex number (for R, Z is real). We can therefore make series and parallel of different components.

- For resistors, voltage and current are in phase (they oscillate with same frequency and no phase shift);
- For capacitors, current and voltage rotate at the same speed, but the first anticipates the second by 90°, i.e. they are in quadrature;
- For inductors, current and voltage rotate at the same speed, but the second anticipates the first by 90°, i.e. they are in quadrature.

Basic Components

Diode



A **non-linear device** with an exponential I-V relation

$$I = I_0 \cdot [e^{(qV/kT)} - 1]$$

I_0 is the Reverse Leakage Current

k is the Boltzmann constant

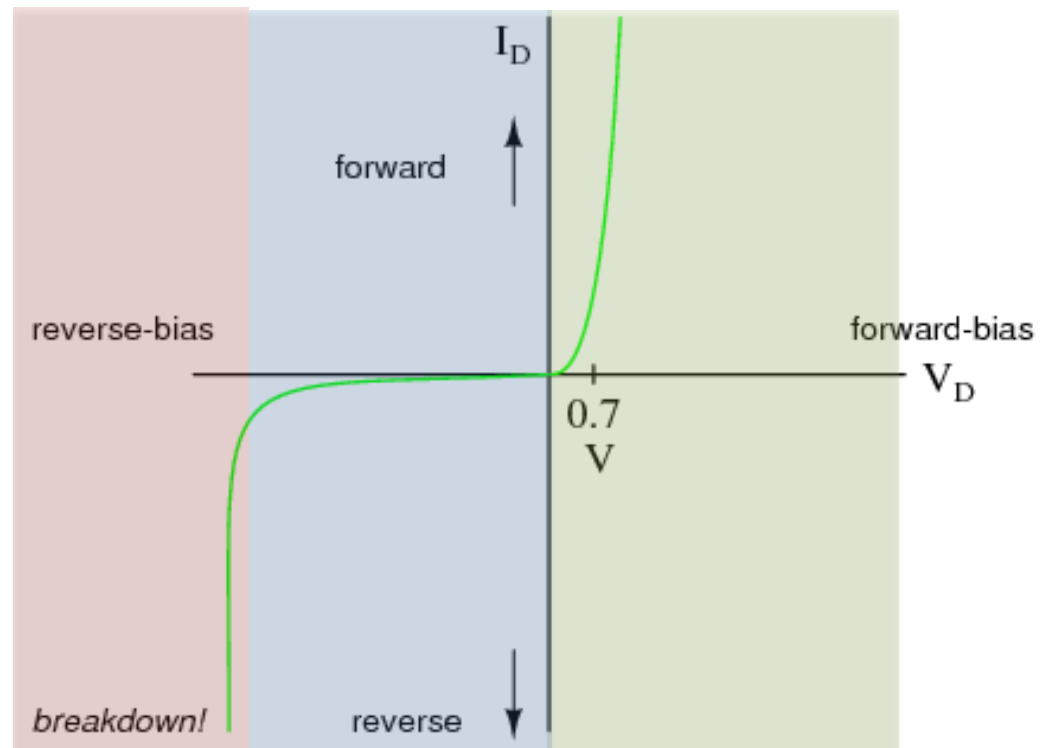
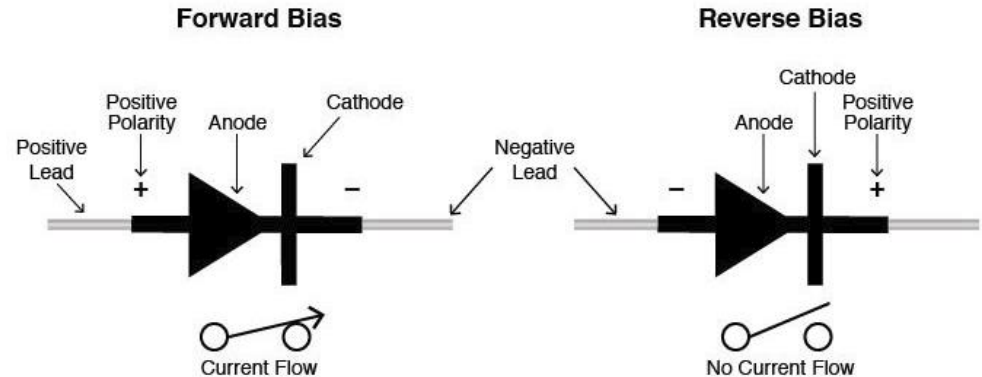
T is the Absolute Temperature

Particular kinds of Diodes:

Zener Diode

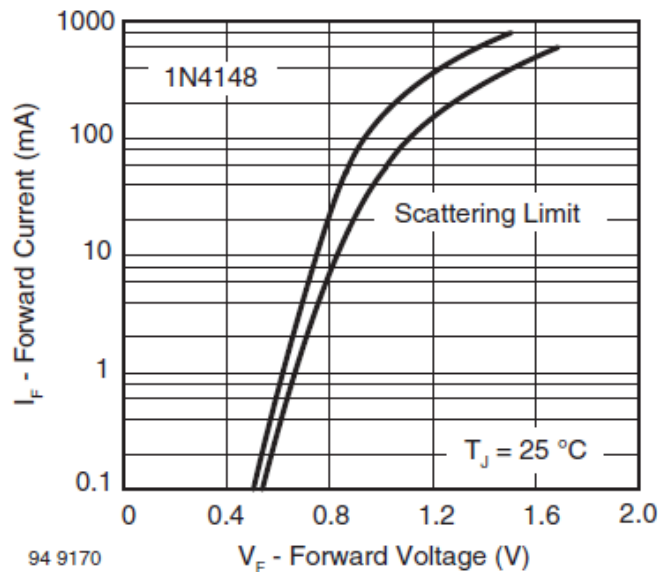
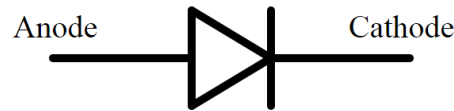
Light Emitting Diode (LED)

Photodiode



Basic Components

Diode



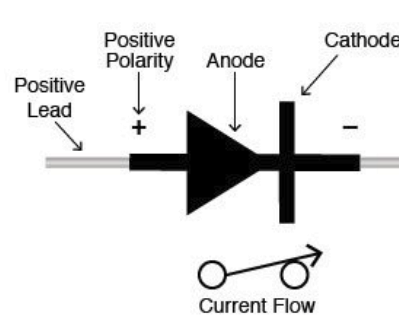
Particular kinds of Diodes:

Zener Diode

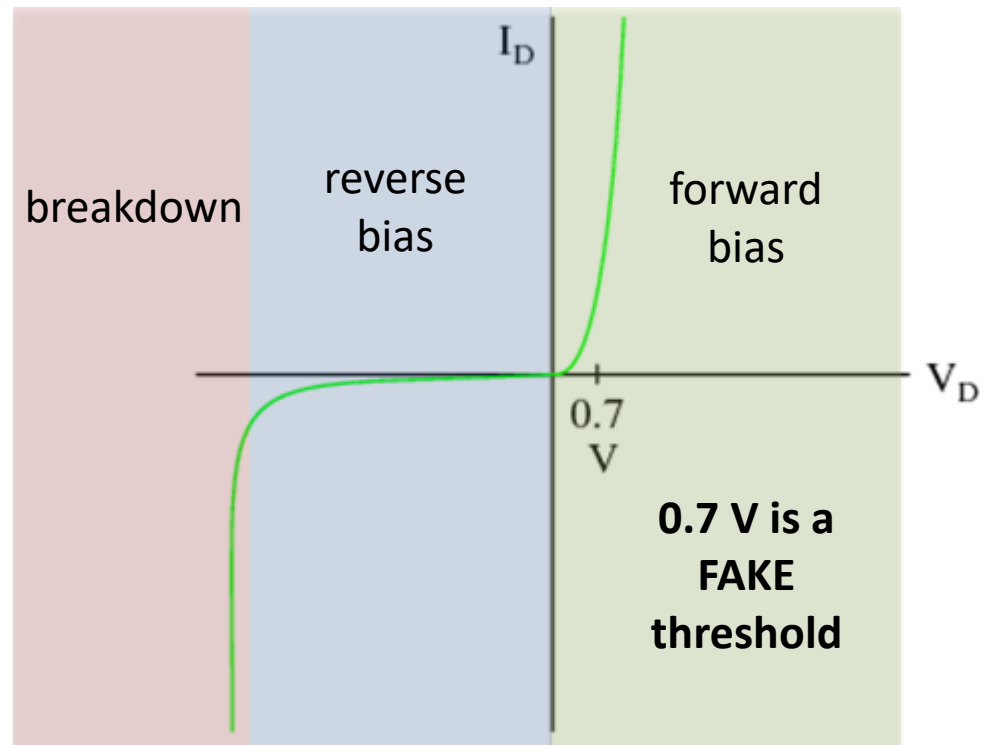
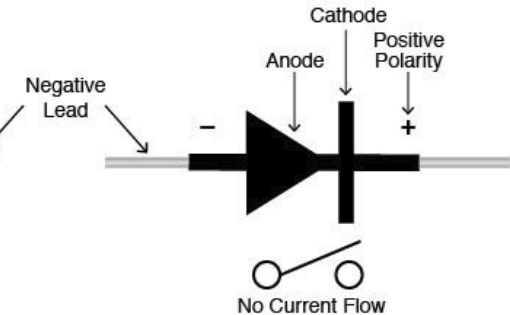
Light Emitting Diode (LED)

Photodiode

Forward Bias

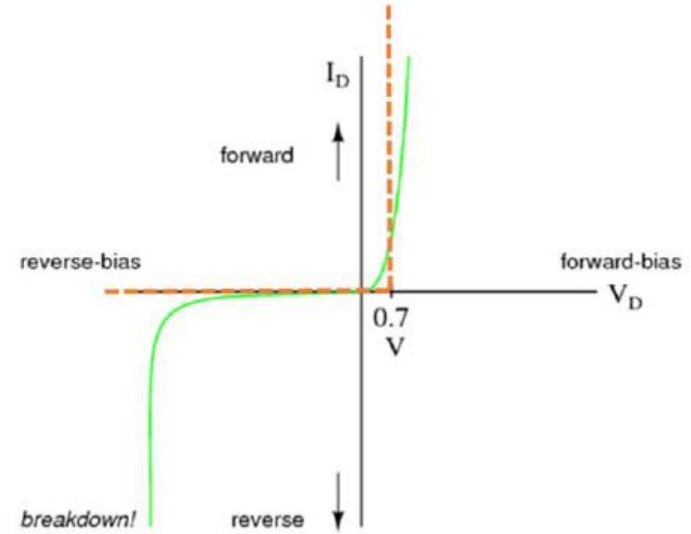
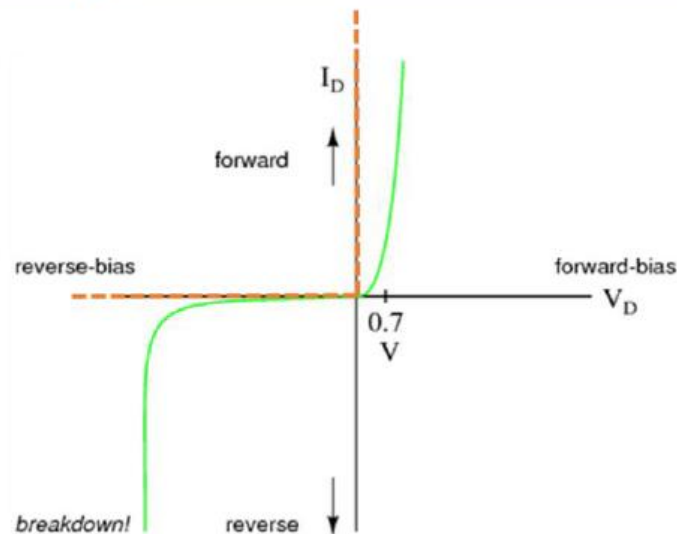
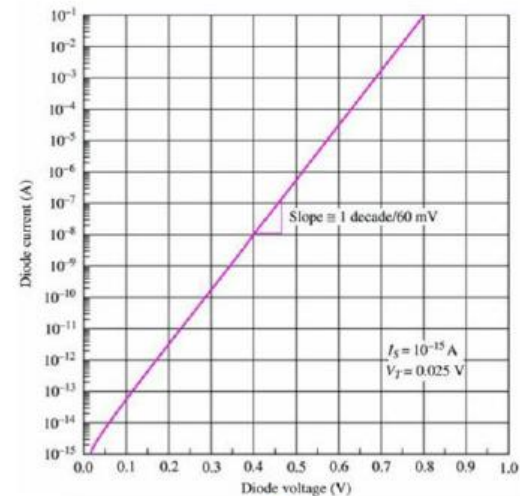
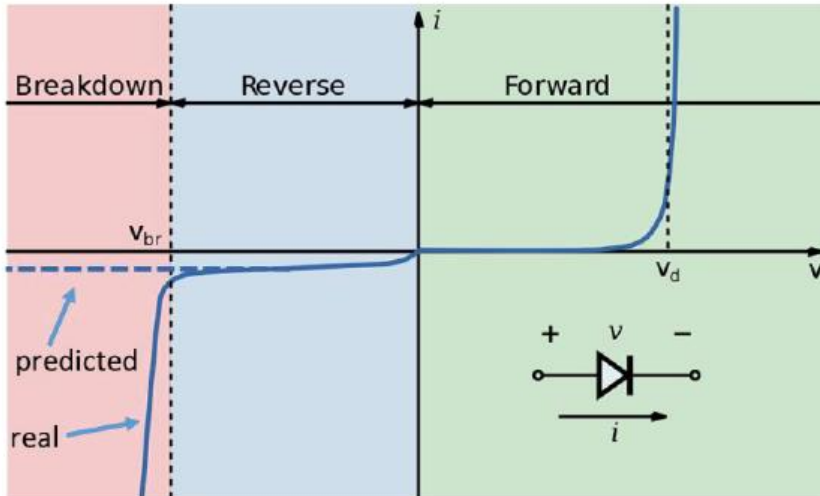


Reverse Bias



Basic Components

Diode Approximate Models



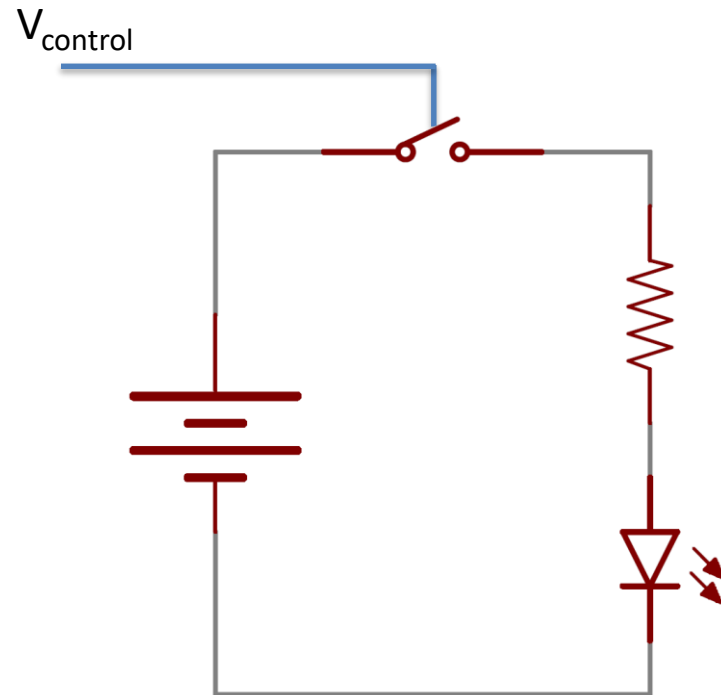
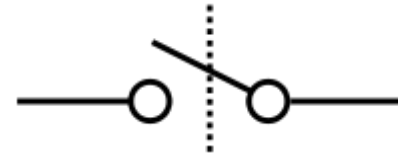
Basic Components

Switches

An electrically-controlled switch is a **three-terminal** device (typically realized by means of a transistor).

Two terminals are connected **in series with the circuit branch** in which current control wants to be attained.

The third terminal is used as a **control** terminal. Depending on the voltage (or the current) at this terminal, the device behaves like a **short or open circuit between the first two terminals**.



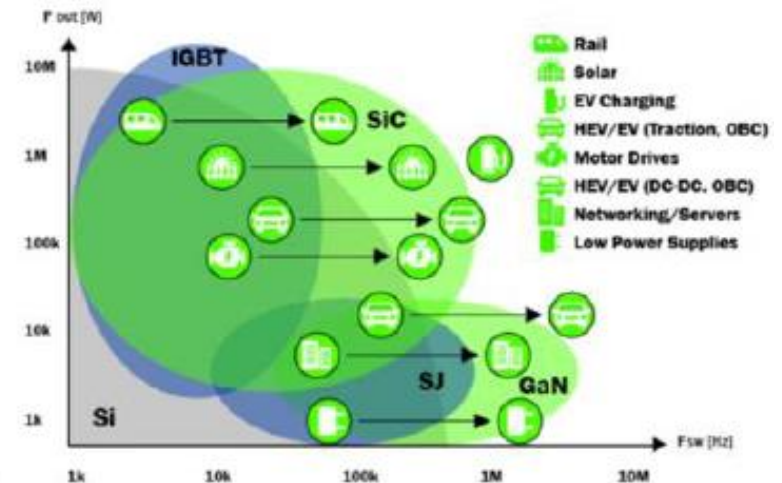
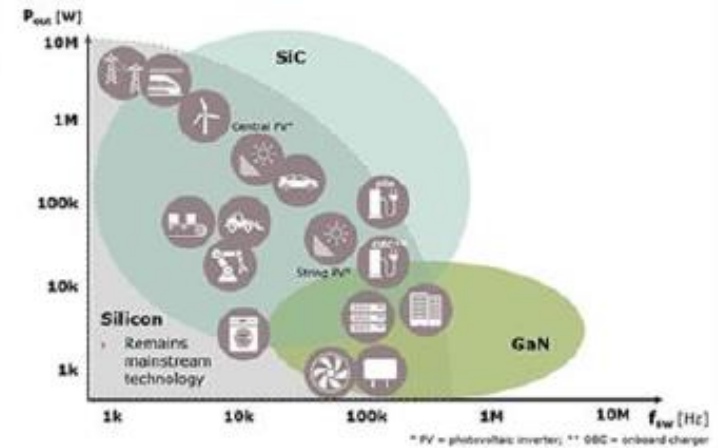
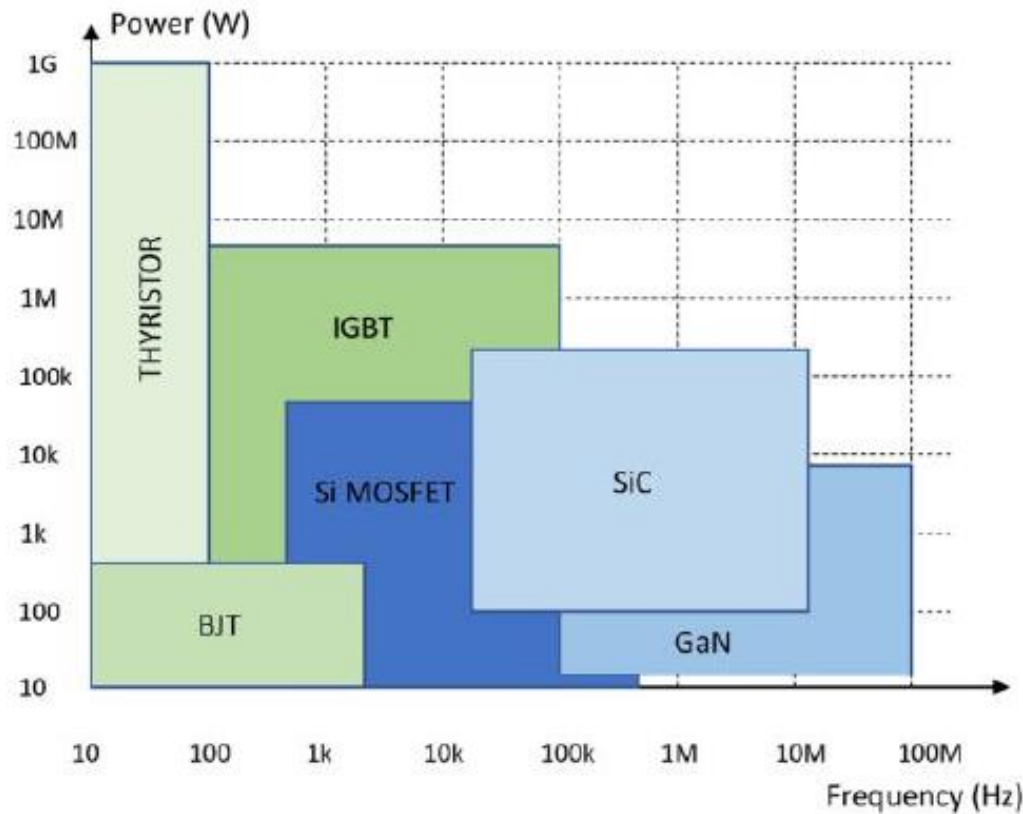
Solid-State Switches

Typologies

- Depending on the current and voltage levels, we use different devices to realize the switch function.
 - **BJT and MOSFET**
 - BJT stands for Bipolar Junction Transistor
 - MOSFET stands for Metal-Oxide-Semiconductor Field-Effect Transistor
 - Low voltages and currents, completely controllable by control signals
 - **IGBT**
 - Insulated Gate Bipolar Transistor
 - Medium voltages (up to 1kV) and high currents (up to 200-400 A) and high frequency
 - **SCR**
 - Silicon Controlled Rectifiers
 - High voltages and currents, switch to ON regulated by a control signal, switch to OFF by power supply
- We will focus here on BJT and MOSFET devices.
- However, recently, devices are evolving from Si toward new materials...

Solid-State Switches

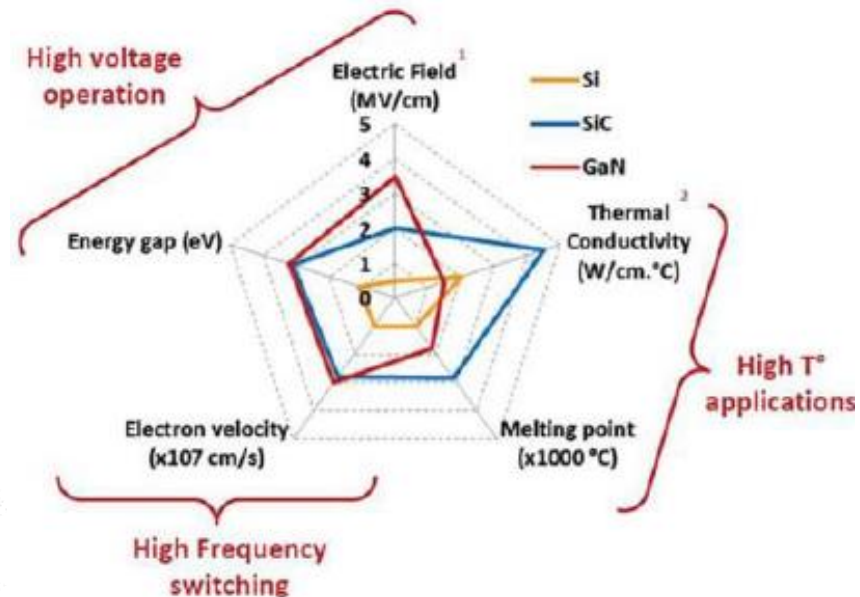
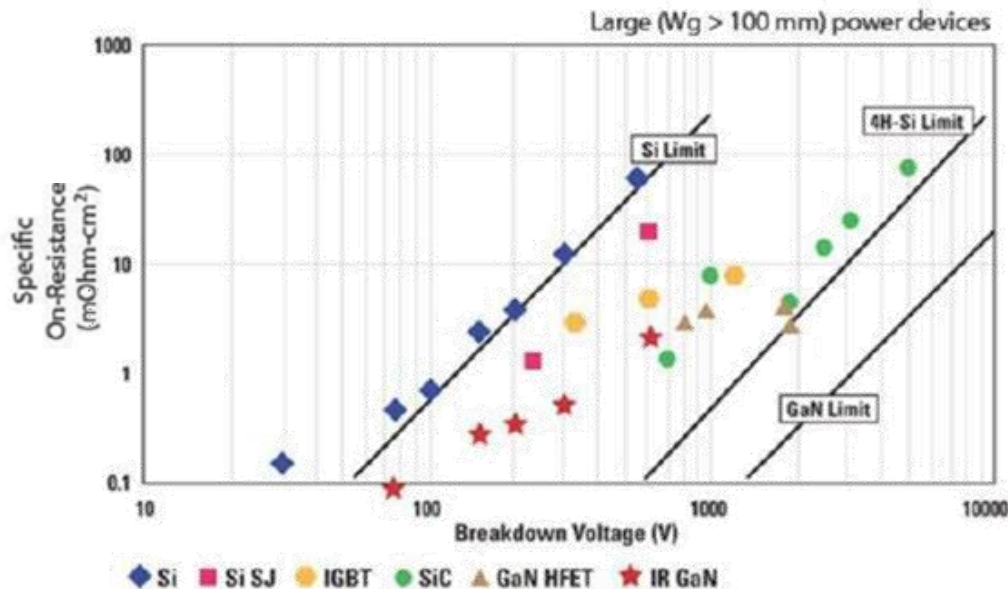
SiC and GaN Power Devices



Solid-State Switches

A Matter of Materials

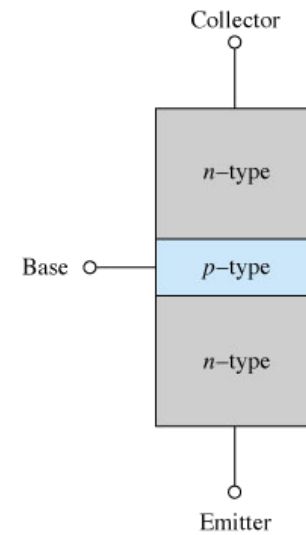
- SiC and GaN have better **Baliga** Figure of Merit that specifies the R_{ON} vs. V_{BD} trade-off: “By relating the device power dissipation to the intrinsic material parameters, [...] power loss can be reduced by using semiconductors with larger mobility and critical electric field for breakdown.”
- Si, SiC, and GaN offer now a varied portfolio of options for different applications



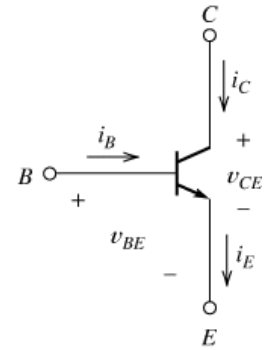
BJT

Bipolar Transistors

- Can be npn or pnp
- Three terminals: Base (B), Collector (C), Emitter (E)
- Can work as an amplifier or as a switch
- Conventional signs for currents (npn case)
 - I_B positive in, I_C positive in, I_E positive out



(a) Simplified physical structure



(b) Circuit symbol

- Large variety of currents (1 mA – ~50 A), voltages (1-100 V), powers (1 mW - ~100 W) and frequencies: different size and packages depending on the maximum power rating
- The maximum power rating defines the amount of heat that the device can withstand (and dissipate) typically with some cooling systems

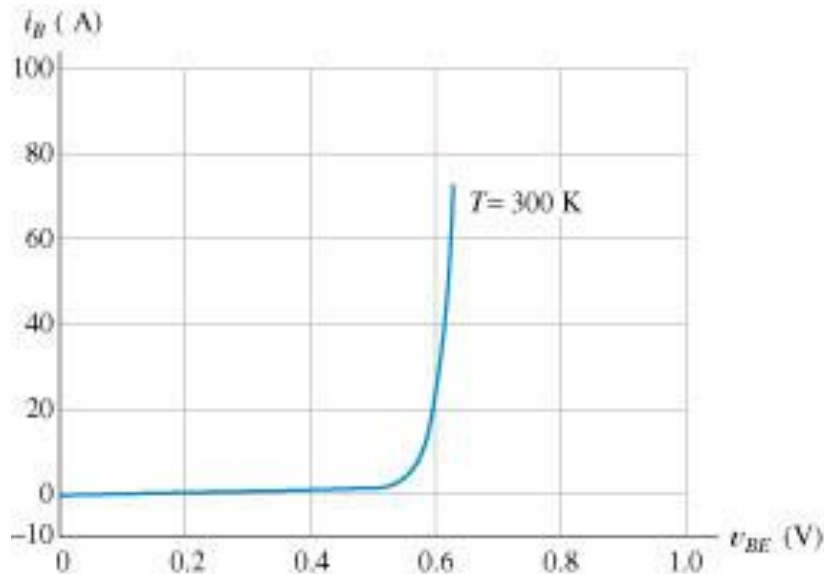
Number	Material	Polarity	Power rating	Voltage rating (Vce)	Current rating	Frequency (MHz)
2N2222	S	N	0.5 W	30	150 mA	250
2N2907	S	P	1.8 W	40	150 mA	200
2N2928	G	P	150 mW	13	2 mA	400
2N6308	S	N	125 W	360	3 A	5
2N6280	S	N	250 W	140	20 A	30

BJT

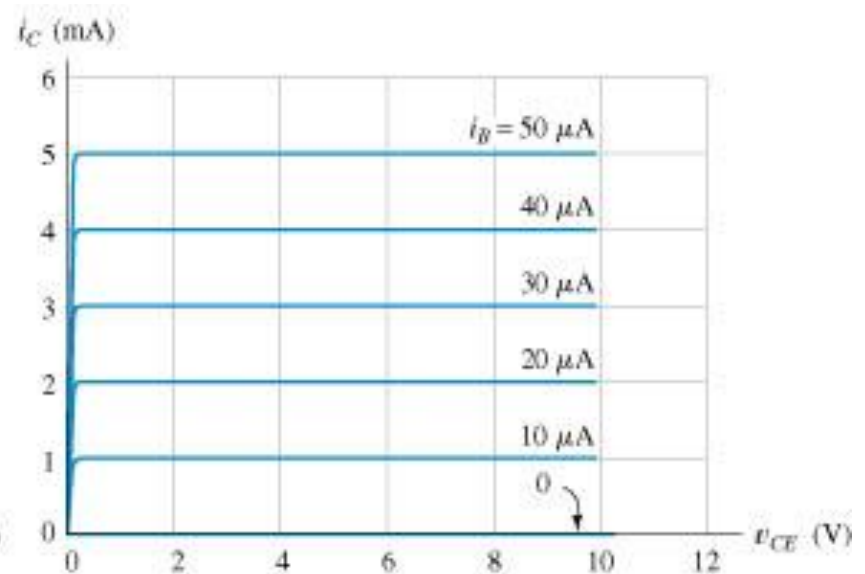
I-V curves

I_B - V_{BE} characteristics and I_C - V_{CE} characteristics

- β_F is the static current gain in common emitter mode with typical value around 100
- pnp and npn static curves look the same except for a flip in the voltage signs



(a) Input characteristic



(b) Output characteristics

BJT

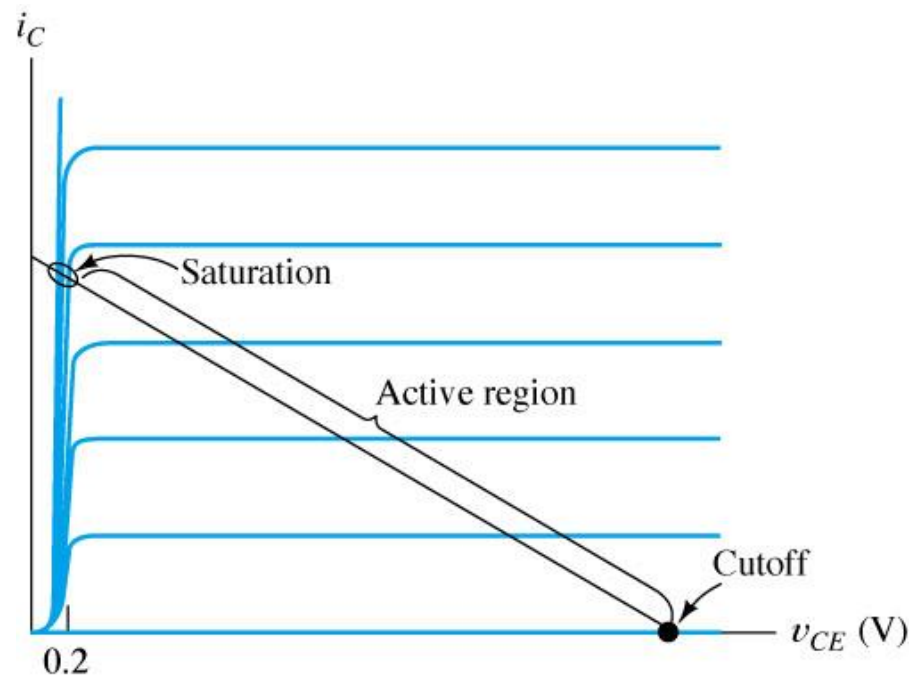
Regions

BJT can work in 3 regions:

Active Region: $V_{BE} \approx 0.7 \text{ V}$; $V_{CE} > 0.3 \text{ V}$; $I_C = \beta_F \cdot I_B$; $I_B > 0$.

Saturation: $V_{BE} \approx 0.7 \text{ V}$; $V_{CE} \approx 0.3 \text{ V}$; $I_C < \beta_F \cdot I_B$; $I_B > 0$.

Cut-off: $V_{BE} < 0.5 \text{ V}$; $V_{BC} < 0.5 \text{ V}$; $I_C = 0$; $I_B = 0$.



BJT

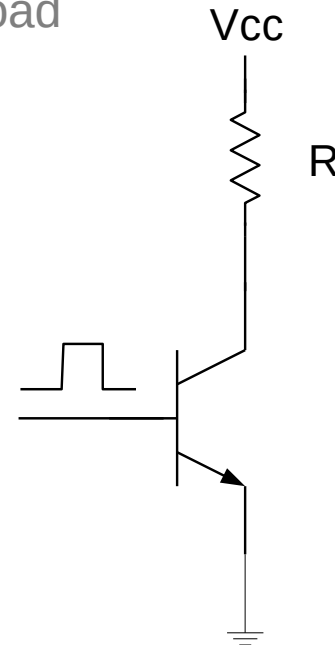
Use as a switch

It acts like a digital device in the sense that has only two states: ON and OFF.
Controls the power at its output

In OFF (cut-off mode), V_{CE} is high (can be as high as V_{CC}), $I_C=0$ and no power dissipation

In ON (saturation), the BJT is fully on, V_{CE} is low (around 0.3 V) and I_C is big so the BJT dissipates power and drives power to the load

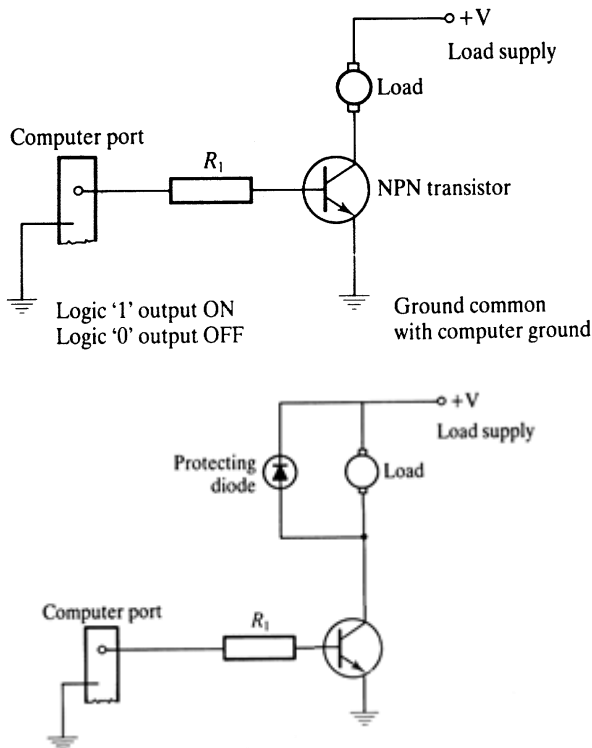
Dissipated power turns into heat!



BJT

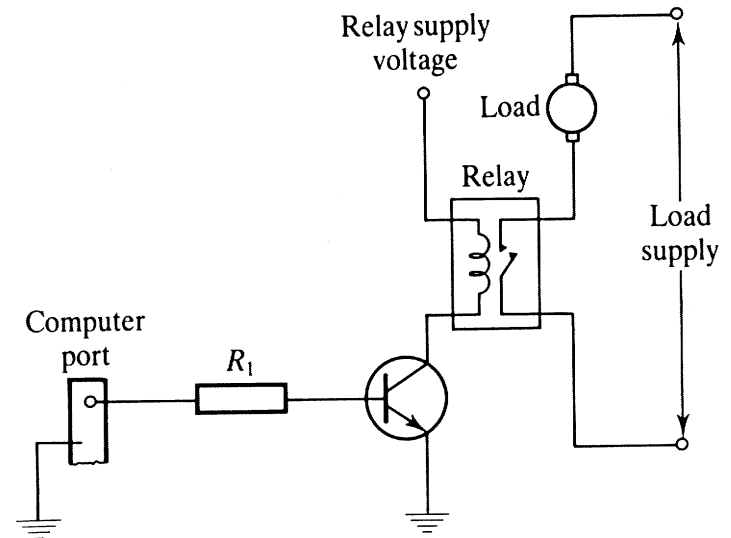
Use as switch

- BJT as a switch
- ON: output voltage is $V_{CEsat} \approx 0.2$ or $0.3V$ (Low - saturation)
- OFF: output voltage is $+5V$ (high - cutoff)



Average power control can be achieved by means of a special technique:
Pulse Width Modulation

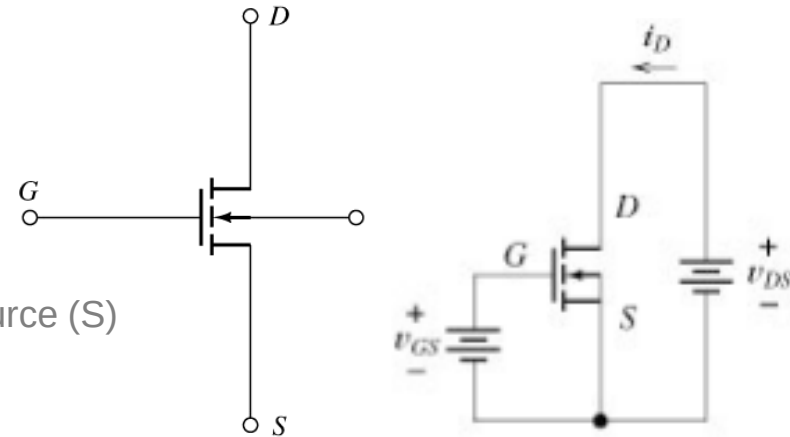
In case the BJT cannot stand the load current a relay can be used



MOSFET

Field-Effect Transistors

- Can be n-type or p-type
- Four terminals: Gate (G), Body or Bulk (B), Drain (D), Source (S)
- Can work as an amplifier or as a switch
- Insulated gate implies no gate current
- Bulk typically connected to the source (not always)
- Drain and Source are symmetrical (Drain is the one at a higher voltage)
- By applying a large enough V_{GS} , a channel is formed between D and S
 - In this case, applying a V_{DS} , current can flow between D and S

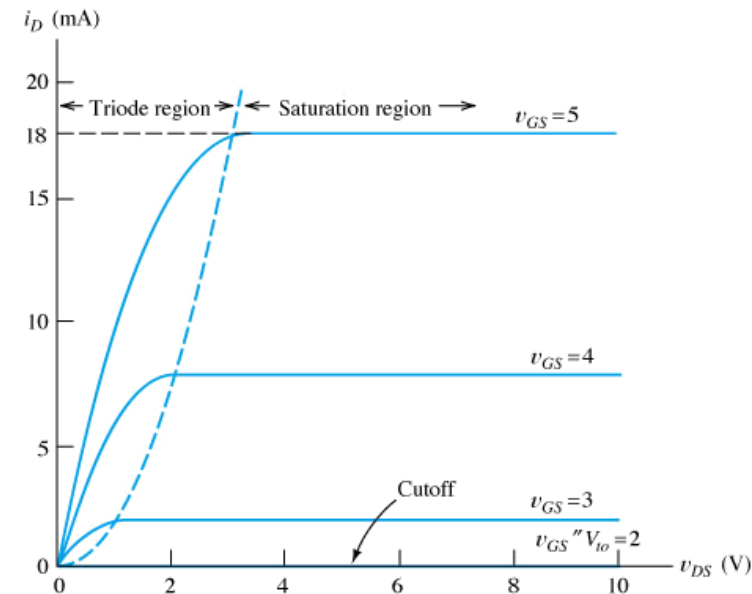


- 3 working regions:

Saturation: $V_{GS} > V_{TH}$; $V_{DS} > V_{GS} - V_{TH}$; $I_D > 0$ (no V_{DS} dep.)

Triode: $V_{GS} > V_{TH}$; $V_{DS} < V_{GS} - V_{TH}$; $I_D > 0$ (V_{DS} dep.)

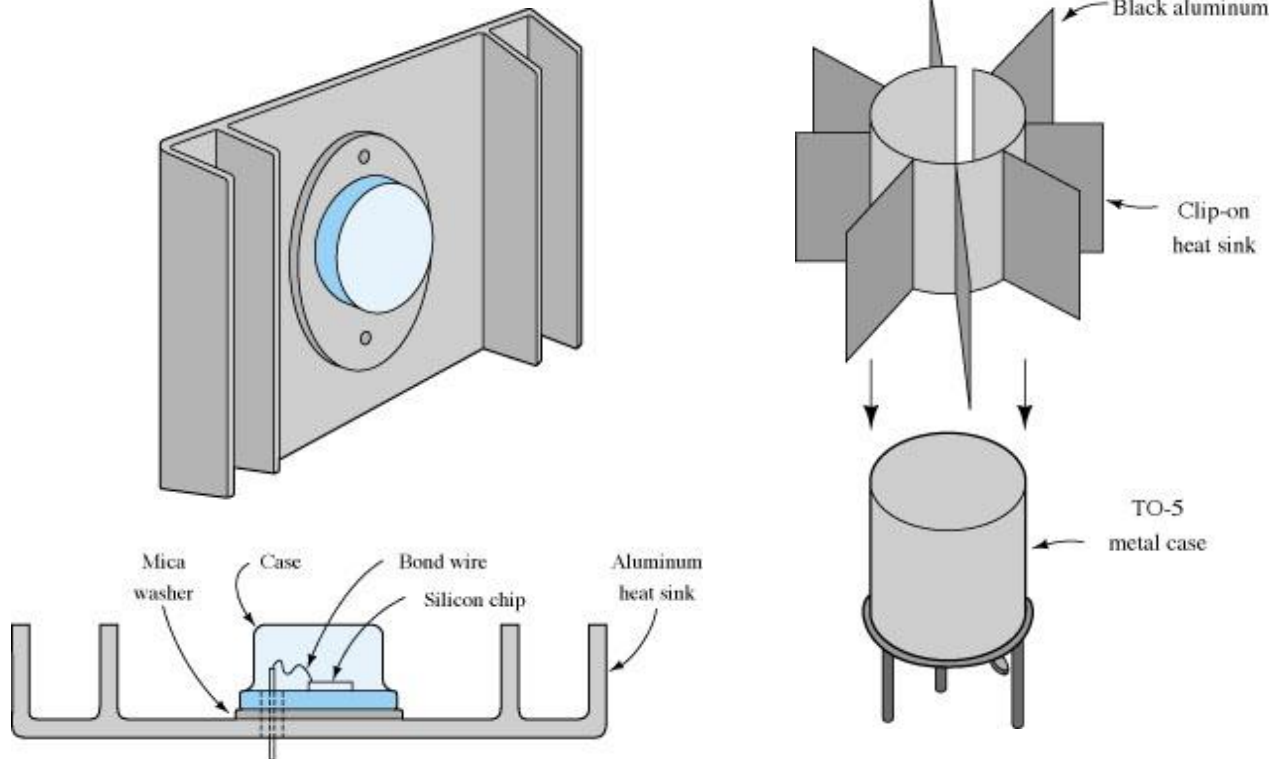
Cut-off: $V_{GS} < V_{TH}$; $I_D = 0$.



Power Devices

Handling Power

- Amplifiers and power supply apparatus dissipate power
- Must design appropriate cooling systems to avoid overheating



Power Devices

Handling Power

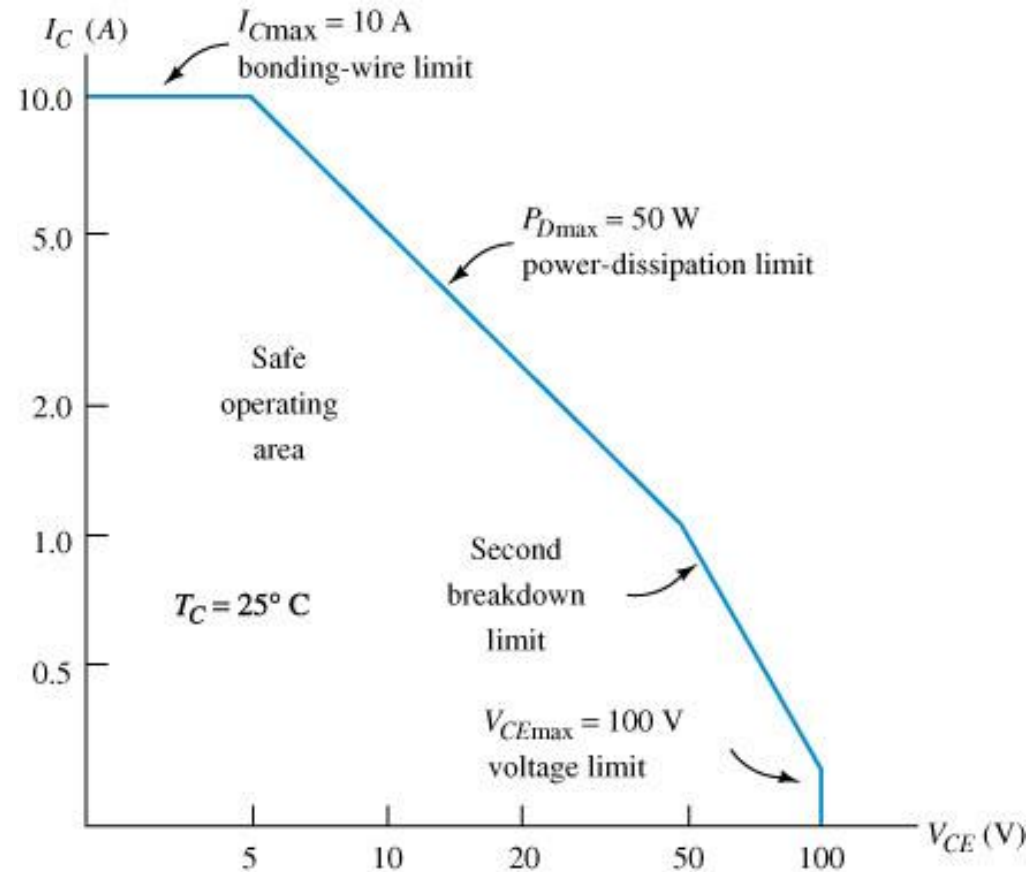
- BJT and MOSFET can be grouped into
 - Signal devices
 - Power devices
- Power devices have typically:
 - Lower gain
 - Higher capacitance
 - Lower corner frequencies
 - Larger leakage

	Signal BJT (2N222A)	Power BJT (2N3055A)
P_{Dmax} (W) @ 25°C	1.8	115
I_{Cmax} (A)	0.8	15
V_{CEmax} (V)	40	60
β_{min}	35-100	5-20
$V_{BE_{max}}$ (V)	6	7
f_t (MHz)	300	0.8
C_{obo} (pF)	8	60-600

Power Devices

Handling Power

- BJT and MOSFET have a Safe Operating Area (SOA)
 - Signal devices
 - Power devices
- BJT suffers from thermal runaway
 - Condition in which the T increase leads to an I increase, which in turn increases P and T
 - Positive feedback that can bring the system to thermal destruction
- MOSFET has no such issue
- MOSFETs with double diffused drain have high breakdown voltages, and low series resistance



- **BJT needs a large I_B to drive large I_C (perhaps too large an I_B) and suffer thermal runaway. Also, input impedance is low**
- **MOSFETs are preferred**

Op-Amp

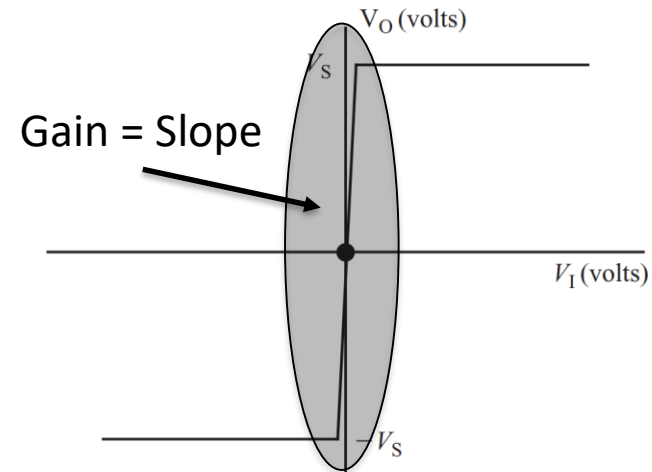
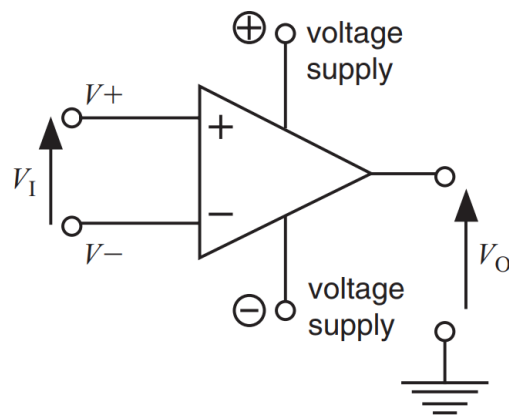
What is it?

Basically, a **very good differential amplifier**!

Its aim is to **respond only to the difference** between its input voltage terminals (i.e. **differential** mode input) and **not to the actual values** of either inputs (i.e. **common** mode input).

It is an **active** circuit, i.e., **it needs power supply**. It provides **amplification** (ideally only) of the differential input by a **very large factor**, called the **gain**, typically in the range of 10^6 to 10^7 and beyond for very high gain topologies. Ideally it should be infinite. Its output voltage is though limited by the supply voltage!

Sounds strange! It seems like a **total waste** to have a huge gain and then being limited to a very narrow input voltage range before reaching saturation!!!



Op-Amp Comparator

One application in which this strangeness disappears is the comparator!

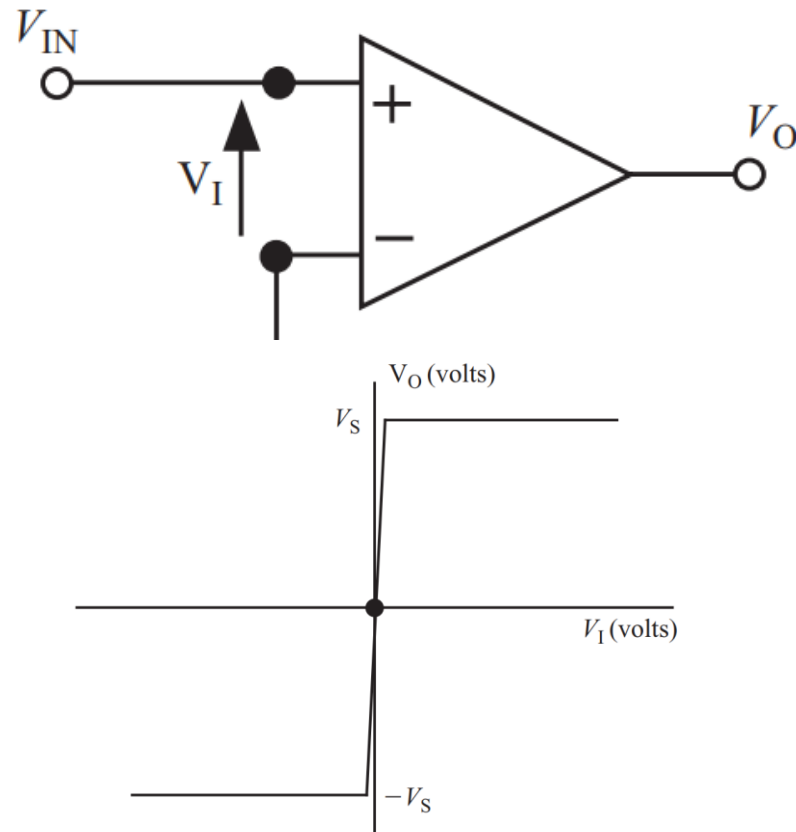
Compares the input voltage to a reference voltage.

It is sufficient for the input voltage to be infinitesimally (in the ideal case) above or below the reference voltage for the op-amp to saturate to either $+V_{ss}$ or $-V_{ss}$

In the typical real case, being $\sim 100 \mu\text{V}$ above or below the reference voltage is sufficient to achieve saturation.

This mechanism is at the basis of the A/D conversion (we will take this into account later on).

However, what if I want to use it as an actual amplifier???



Op-Amp

Features of the Ideal Op-Amp

The ideal op-amp should display:

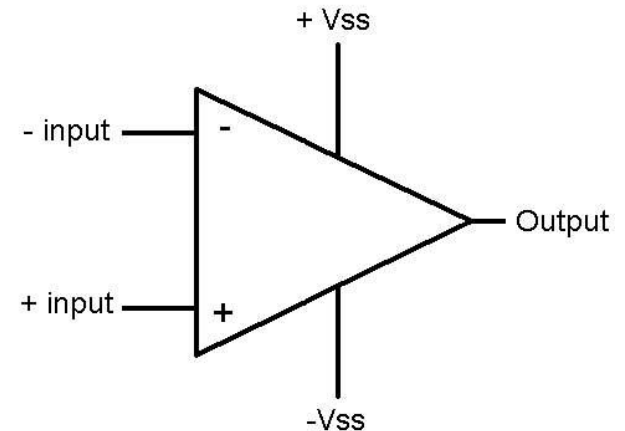
- Zero input currents (bias current)
- Infinite Gain
- Infinite Bandwidth
- Infinite Slew Rate

Very easy to realize negative (and also positive) feedback!

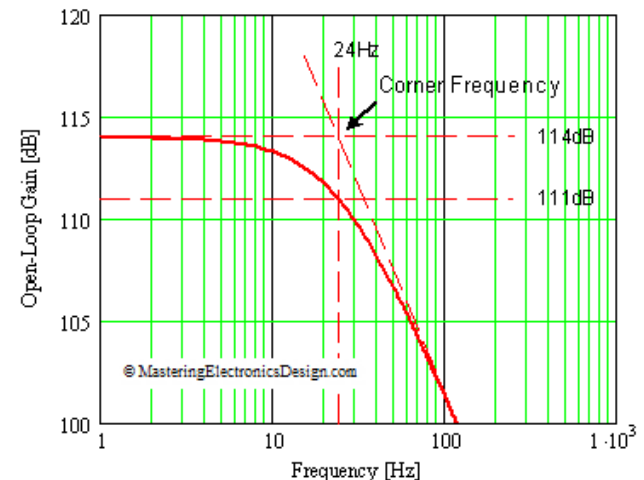
A real op-amp (see the picture) has a large but finite gain

Also, a very narrow bandwidth!!!

How is that???



Analog Devices ADA4004



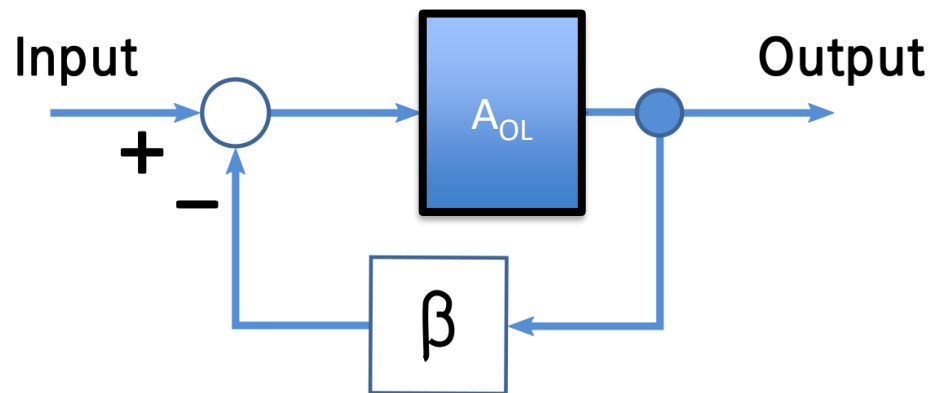
Op-Amp

Negative Feedback

The answer to all previous doubts is **Negative Feedback!**

Thanks to NF, we achieve a lot of advantages at a relatively small cost.

It is also the way to get the **virtual short-circuit property!**



Provided a very large open-loop gain (actually, a very large loop gain) we have:

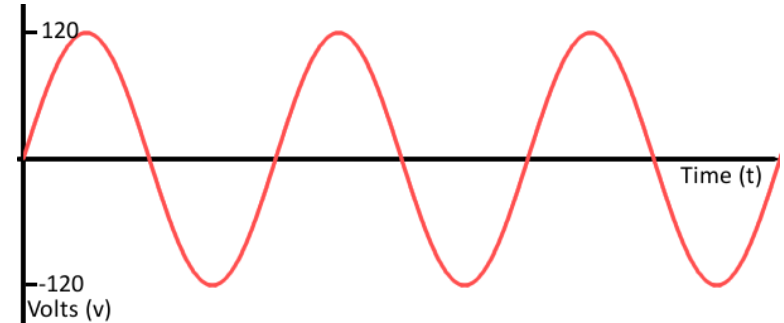
- Closed-loop gain is not dependent anymore on the open-loop one (or at least very weakly).
- Worse gain than in open-loop
- Larger bandwidth than in open-loop
- Better I/O impedances than in open-loop

Analog vs. Digital

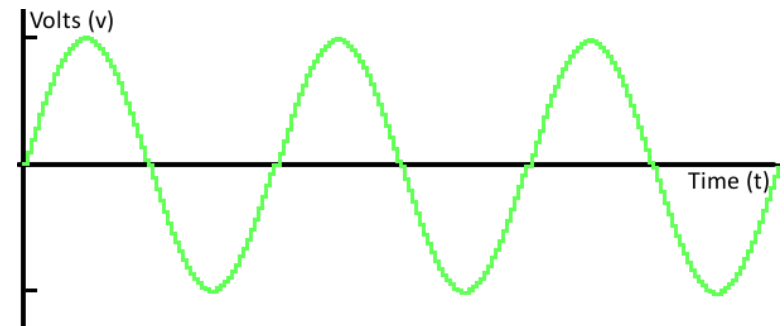
Another Tale of Two Domains

Because a signal varies over time, it's helpful to plot it on a graph where time is plotted on the horizontal axis, and voltage (or current) on the vertical axis. The graph of an **analog signal** should be smooth and continuous.

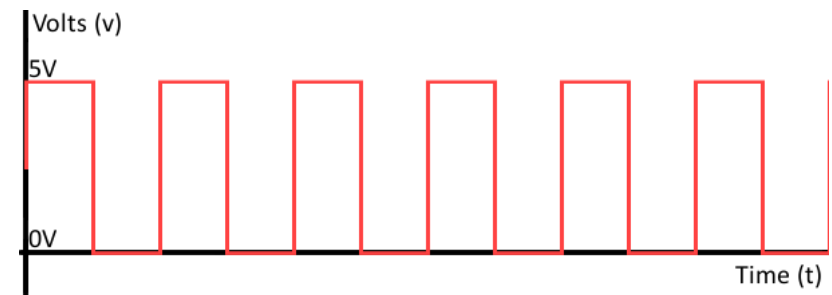
Digital signals must have a finite set of possible values. The number of values in the set can be anywhere between two and an arbitrarily high finite number. Most commonly digital signals will be one of two values, e.g. either 0V or 5V. Timing graphs of **digital signals** look like square waves.



SAMPLING AND QUANTIZATION



DIGITIZATION



Analog vs. Digital

Another Tale of Two Domains

	Analog	Digital
Signal	Analog signal is a continuous signal which represents physical measurements.	Digital signals are discrete time signals generated by digital modulation.
Waves	Denoted by sine waves	Denoted by square waves
Representation	Uses continuous range of values to represent <u>information</u>	Uses discrete or discontinuous values to represent information
Example	Human voice in air, analog electronic devices.	Computers, CDs, DVDs, and other digital electronic devices.
Technology	Analog technology records waveforms as they are.	Samples analog waveforms into a limited set of numbers and records them.
Data transmissions	Subjected to deterioration by noise during transmission and write/read cycle.	Can be noise-immune without deterioration during transmission and write/read cycle.
Response to Noise	More likely to get affected reducing accuracy	Less affected since noise response are analog in nature

Analog vs. Digital

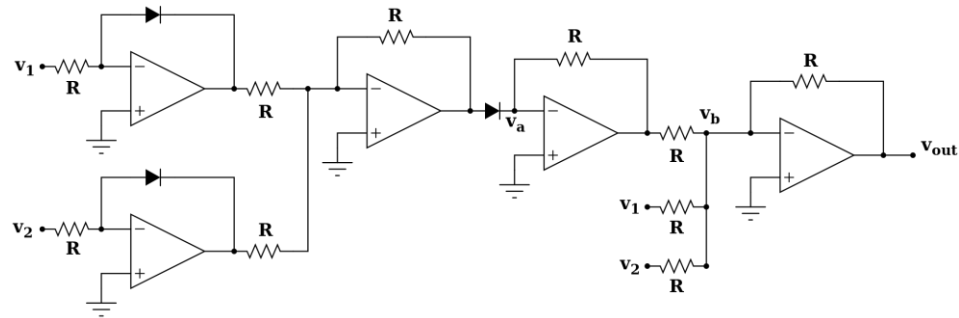
Another Tale of Two Domains

	Analog	Digital
Flexibility	Analog hardware is not flexible.	Digital hardware is flexible in implementation.
Uses	Can be used in analog devices only. Best suited for audio and video transmission.	Best suited for Computing and digital electronics.
Applications	Thermometer	PCs, PDAs
Bandwidth	Analog signal processing can be done in real time and consumes less bandwidth.	There is no guarantee that digital signal processing can be done in real time and consumes more bandwidth to carry out the same information.
Memory	Stored in the form of wave signal	Stored in the form of binary bit
Power	Analog instrument draws large power	Digital instrument draws only negligible power
Cost	Cost is high and not easily portable	Low cost and portable
Impedance	Low	High order of 100 megaohm
Errors	Analog instruments usually have a scale which is cramped at lower end and give considerable observational errors.	Digital instruments are free from observational errors like parallax and approximation errors.

Analog vs. Digital

Another Tale of Two Domains

Our world is **analog** (at least from our macroscopic standpoint). For some tasks, particularly **amplification**, analog represents the most elegant, direct, and accurate way to perform the task.



Our lives are based around **logical** decisions. **Digital** electronics is based on logic circuitry. So, digital electronics lends itself very nicely to solving **real-world logic problems** either independently or through the use of computers. Logic circuits can be **combinatorial** or **sequential**.

